

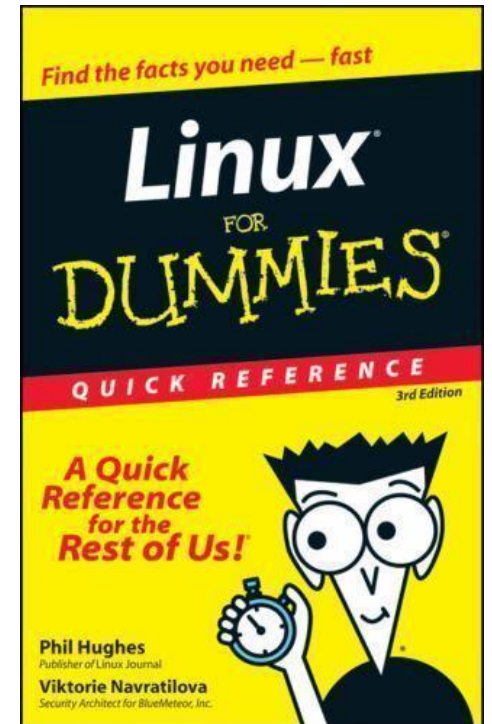
Conservation genomics workshop 2026

Intro to Bash

Mpala, Kenya, 2026

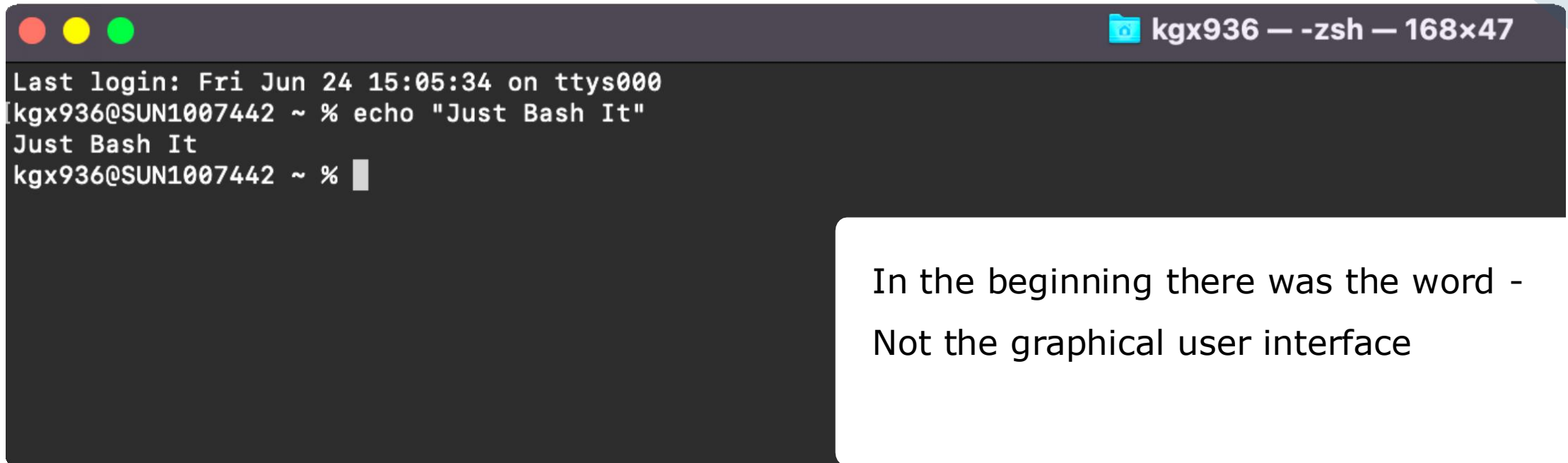
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Learning objective: To become more familiar with the command line



Learning objective: To become more familiar with the command line

1. What is Linux & its use in genetics?
2. UNIX operating system build
3. What is the shell?
4. What is the difference between a graphical and a command line shell?
5. How to navigate and organise files via the command line shell
6. How to write programs and execute them
7. Tutorial

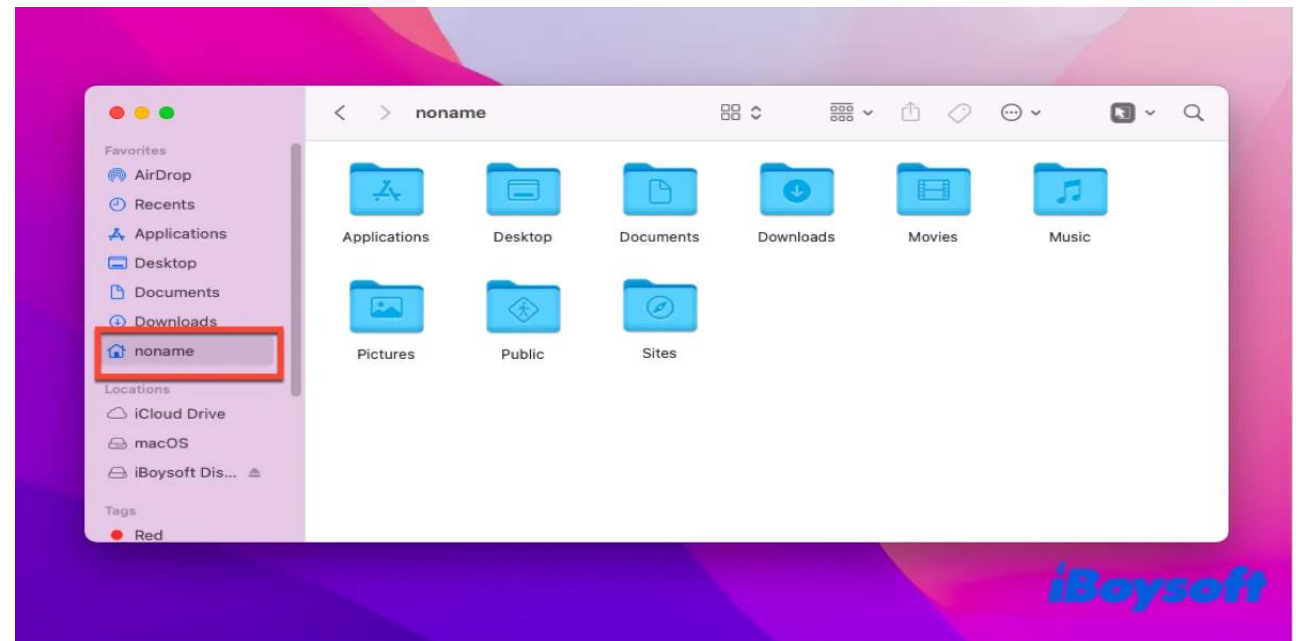
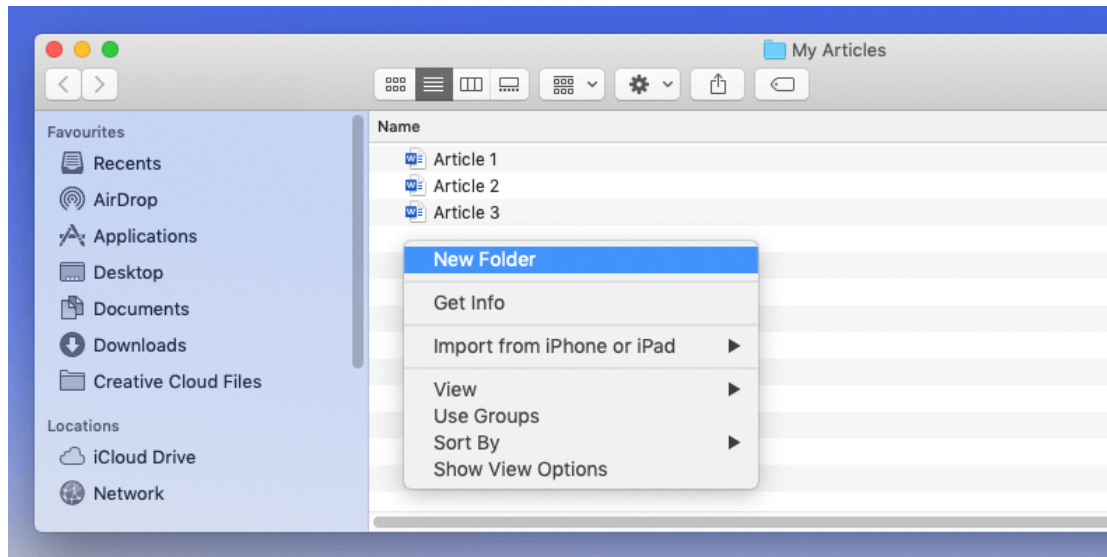
A screenshot of a terminal window with a dark background. The title bar at the top shows three colored window control buttons (red, yellow, green) on the left and a folder icon followed by the text 'kgx936 — -zsh — 168x47' on the right. The terminal content shows a login message 'Last login: Fri Jun 24 15:05:34 on ttys000', a prompt 'kgx936@SUN1007442 ~ %', the command 'echo "Just Bash It"', the output 'Just Bash It', and another prompt 'kgx936@SUN1007442 ~ %' with a cursor. A white callout box on the right side of the terminal contains the text 'In the beginning there was the word - Not the graphical user interface'.

The first computers had no graphical user interfaces (GUI) where you could click. Instead, you interacted with the computer via the terminal.

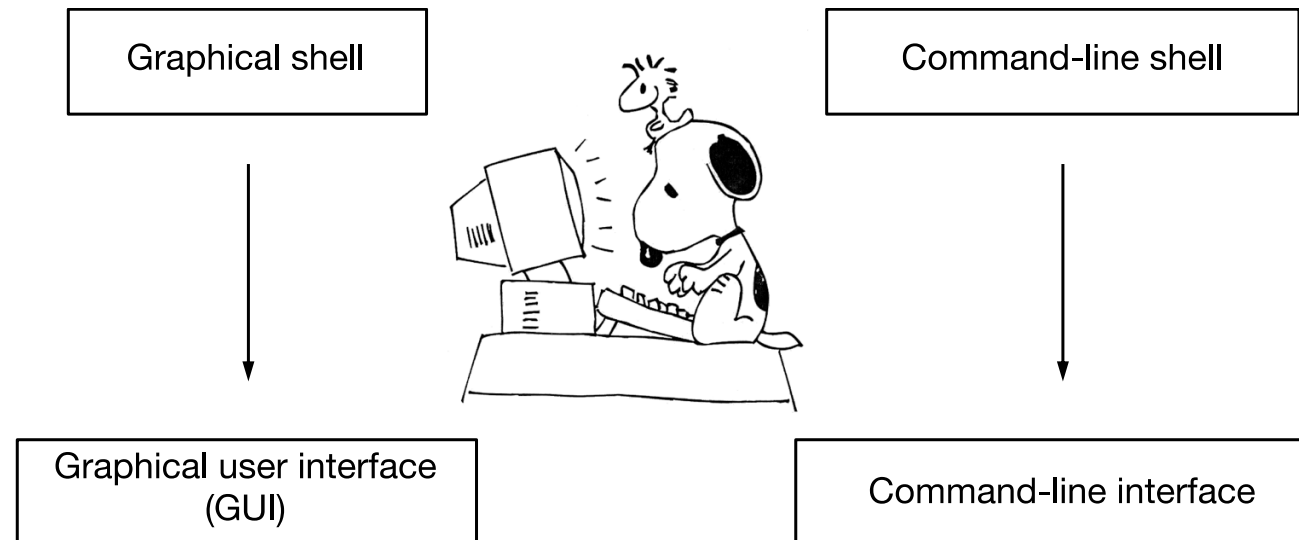
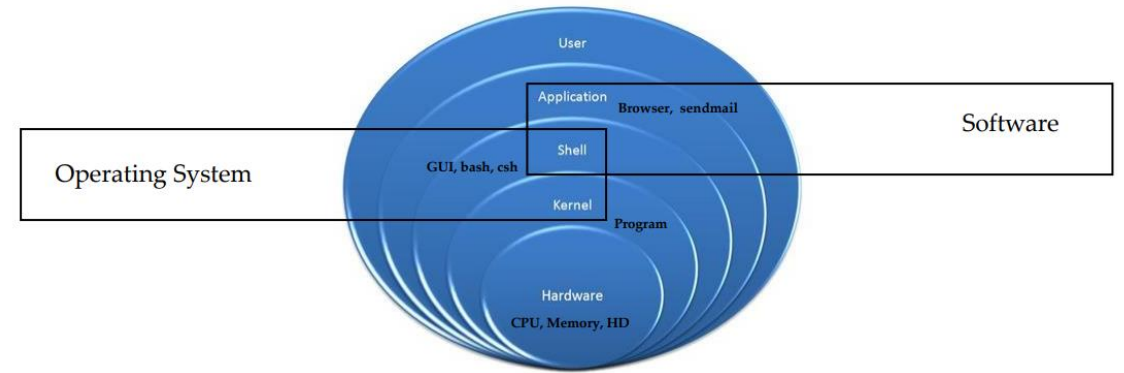
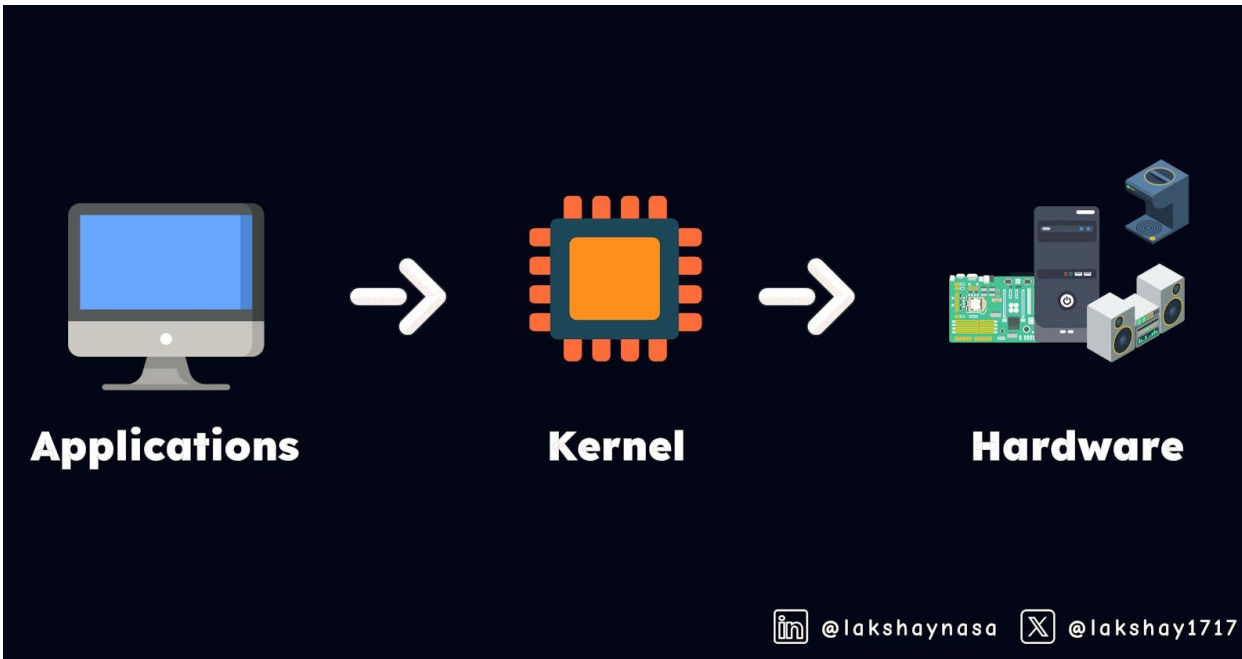
There are still computing environments that do not have GUI.

Some scientific software do not have GUIs either. You need to execute them via the terminal.

Graphical user interface

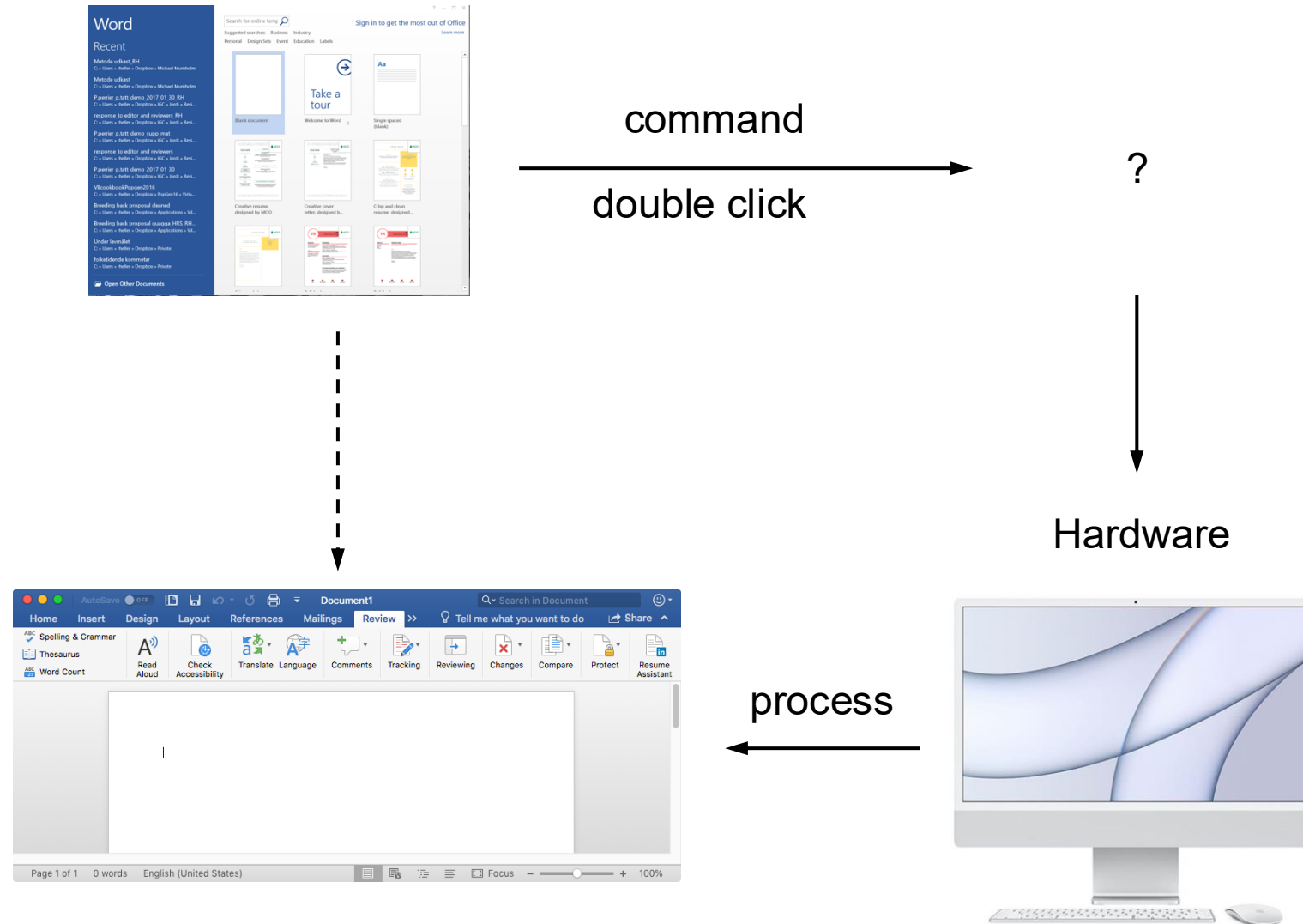


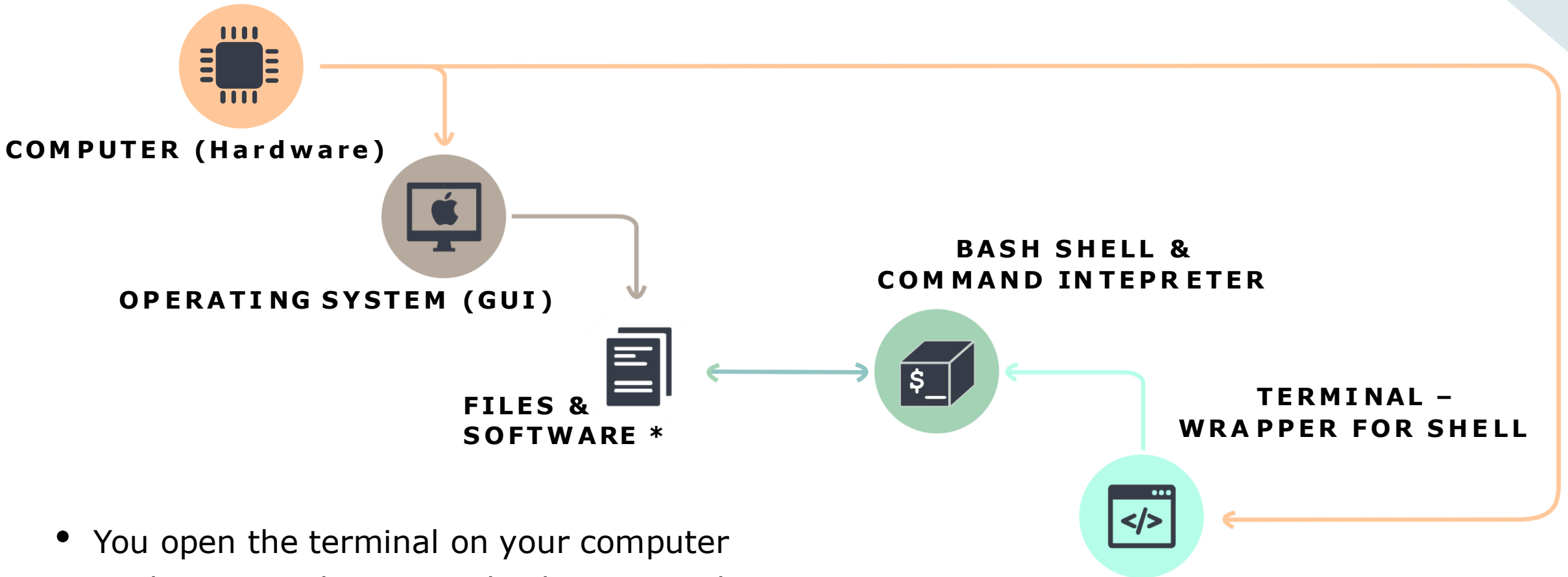
Computers: inside look



Graphical user interface

What happens when you open a document in Word?





- You open the terminal on your computer
- In the terminal you type bash commands
- The shell interprets commands

* software == executable file

Terminal

```
~ — jbf527@scarbpop05fl:/maps/projects/ilab/people/jbf527/2.Reindeer/1.Mapping — zsh — 137x32
...hants/CurrentNe2 — bash  .../jbf527/1.Rawfiles — zsh  ...lityFiltering/output — zsh  ...r/2.QualityFiltering — zsh  ...eindeer/1.Mapping — zsh  ...rse/exercises/linux — zsh  ...es/linux/Advanced — zsh  ...jbf527@a00730:~ — zsh  +

Last login: Fri Jan 17 11:47:50 on ttys000
jbf527@SUN1007694 ~ % ssh jbf527@scarbpop05fl
jbf527@scarbpop05fl's password:

  _ _
 _ _ _
(_)_ _
  0

Welcome to scarbpop05fl
University of Copenhagen
-----
Supported by UCPH IT  it.ku.dk/english

Last login: Tue Jan 21 11:04:22 2025 from 10.203.153.46
[jbf527@scarbpop05fl:~]$ cd /maps/projects/ilab/people/jbf527/
[jbf527@scarbpop05fl:/maps/projects/ilab/people/jbf527]$ ls
0.Scripts  1.Rawfiles  2.Reindeer
[jbf527@scarbpop05fl:/maps/projects/ilab/people/jbf527]$ ll
total 96
drwxr-sr-x. 2 jbf527 comp-prj-ilab  61 Jan 17 07:47 0.Scripts
drwxr-sr-x. 3 jbf527 comp-prj-ilab 1144 Jan 21 12:22 1.Rawfiles
drwxr-sr-x. 4 jbf527 comp-prj-ilab  63 Jan 21 11:00 2.Reindeer
[jbf527@scarbpop05fl:/maps/projects/ilab/people/jbf527]$ cd 1.Rawfiles/
[jbf527@scarbpop05fl:/maps/projects/ilab/people/jbf527/1.Rawfiles]$ ls
ListPublished_B1  PublishedReindeer          SRR11355562_1_fastqc.zip  SRR11355563_2_fastqc.gz  SRR11843135_1_fastqc.zip
ListPublished_B2  SRR11355560_1_fastqc.html  SRR11355562_1_fastqc.gz  SRR11355564_1_fastqc.html  SRR11843135_1_fastqc.gz
ListPublished_B3  SRR11355560_1_fastqc.zip   SRR11355562_2_fastqc.gz  SRR11355564_1_fastqc.zip   SRR11843135_2_fastqc.gz
ListPublished_B4  SRR11355560_1_fastqc.gz    SRR11355563_1_fastqc.html  SRR11355564_1_fastqc.gz    SRR11843136_1_fastqc.gz
ListPublished_B5  SRR11355560_2_fastqc.gz    SRR11355563_1_fastqc.zip  SRR11355564_2_fastqc.gz    SRR11843136_2_fastqc.gz
ListPublished_B6  SRR11355562_1_fastqc.html  SRR11355563_1_fastqc.gz  SRR11843135_1_fastqc.html
[jbf527@scarbpop05fl:/maps/projects/ilab/people/jbf527/1.Rawfiles]$ ll
total 211061064
-rw-r--r--. 1 jbf527 comp-prj-ilab          73 Jan 15 18:36 ListPublished_B1
-rw-r--r--. 1 jbf527 comp-prj-ilab        120 Jan 15 18:37 ListPublished_B2
```

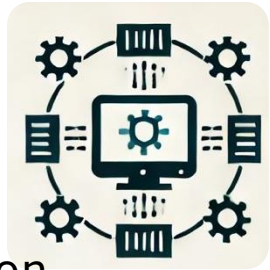
Why should I care?

Bash and command line it's free!

Fast



Flexibility and
versatile

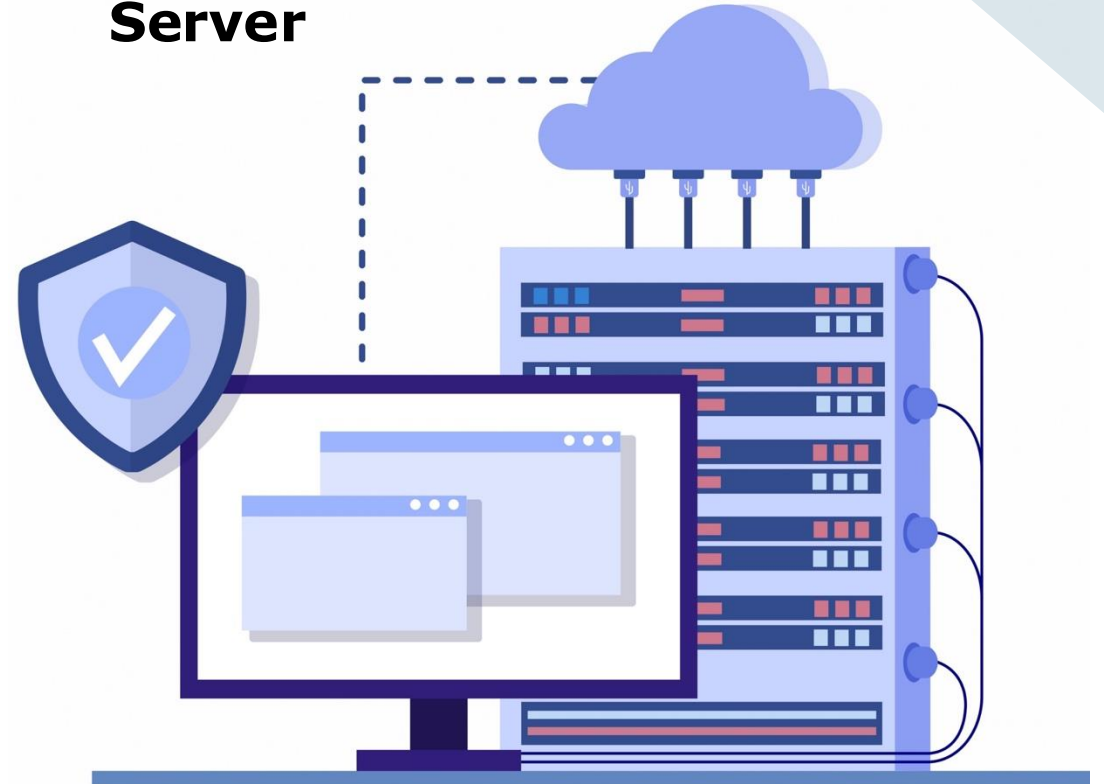


Automation,
reproducible
and parallelised



Big data and
heavy processes

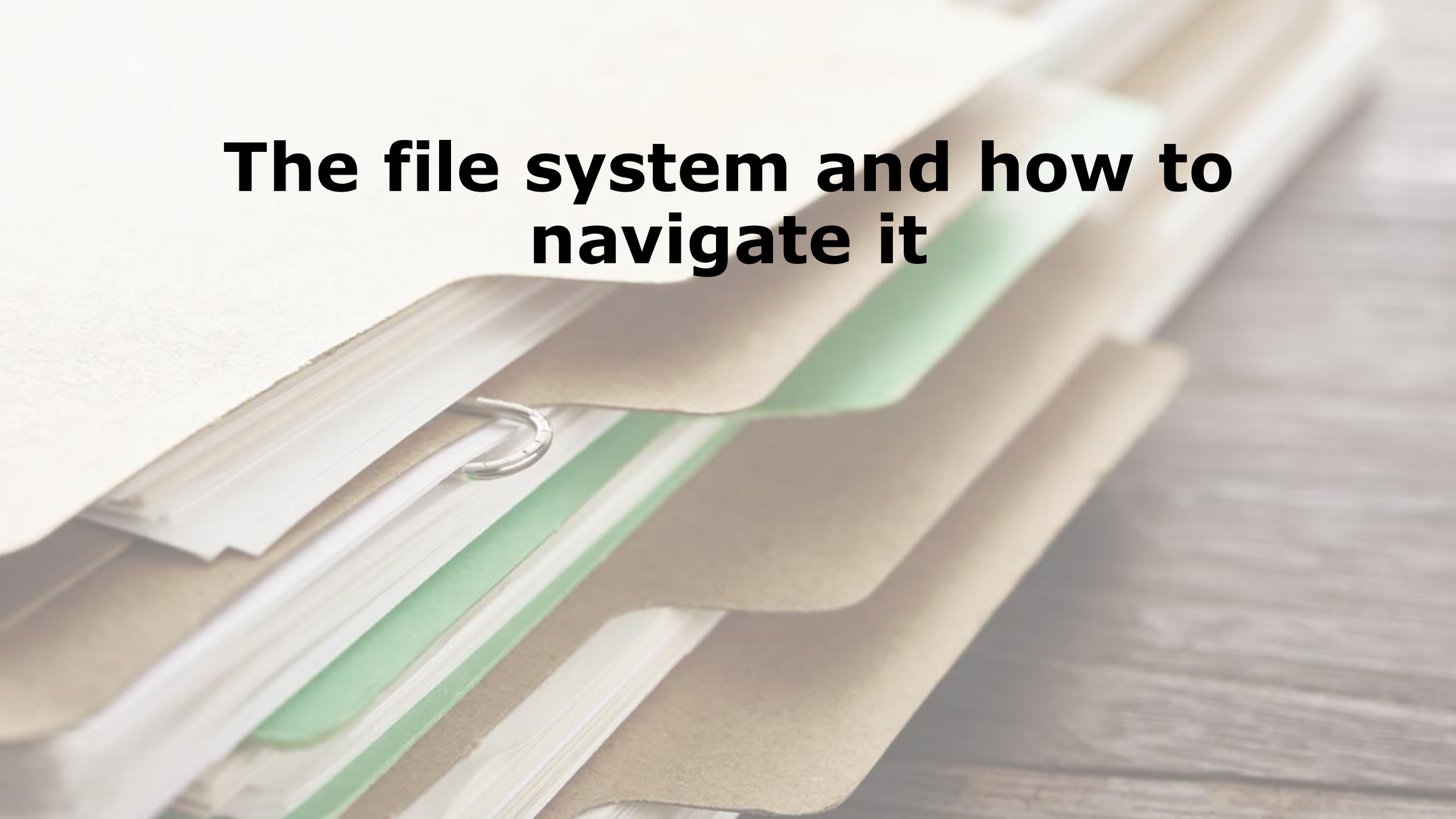
Server



- Processing power
- Remote access
- Storage
- Resource sharing
- Scalability



The file system and how to navigate it



Command line interface: shell

```
jbf527 — nuno@emily: ~ — ssh -Y nuno@emily.popgen.dk — 145x37
...cf — bash  ...opgen.dk  ...opgen.dk  ...] — bash  ...] — bash  ...] — bash  ...] — bash  ...] — bash  ...] — bash  ...g — bash  ...ls — bash  bash  ish  +

bash-3.2$ ssh -Y nuno@emily.popgen.dk
nuno@emily.popgen.dk's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-50-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

24 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

60 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

*** System restart required ***
Last login: Wed Jan 21 14:00:29 2026 from 130.225.188.129
nuno@emily:~$
```


Command line interface: shell

```
[pbr316@popgen-bio groupdirs]$ _
```

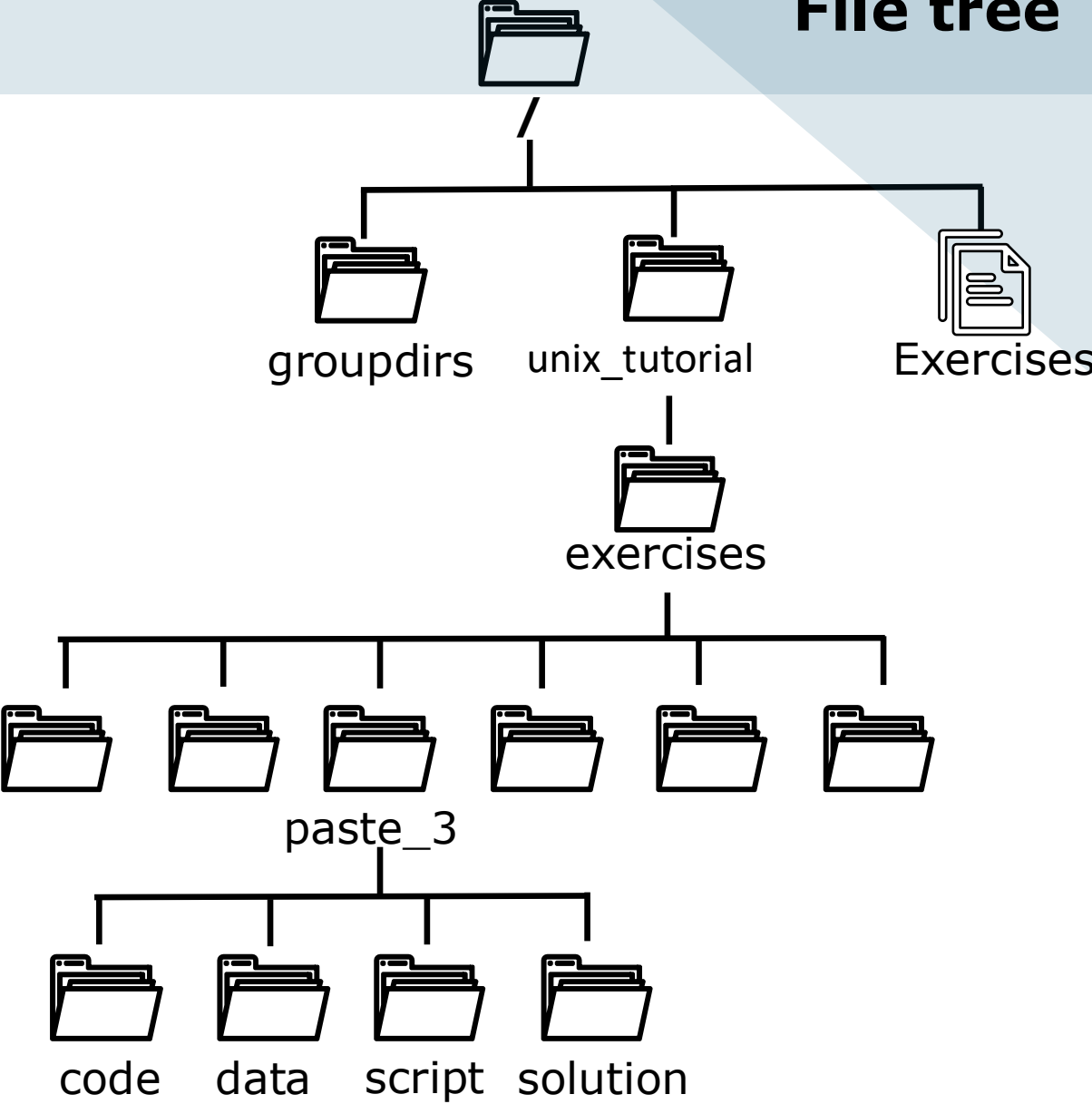
User

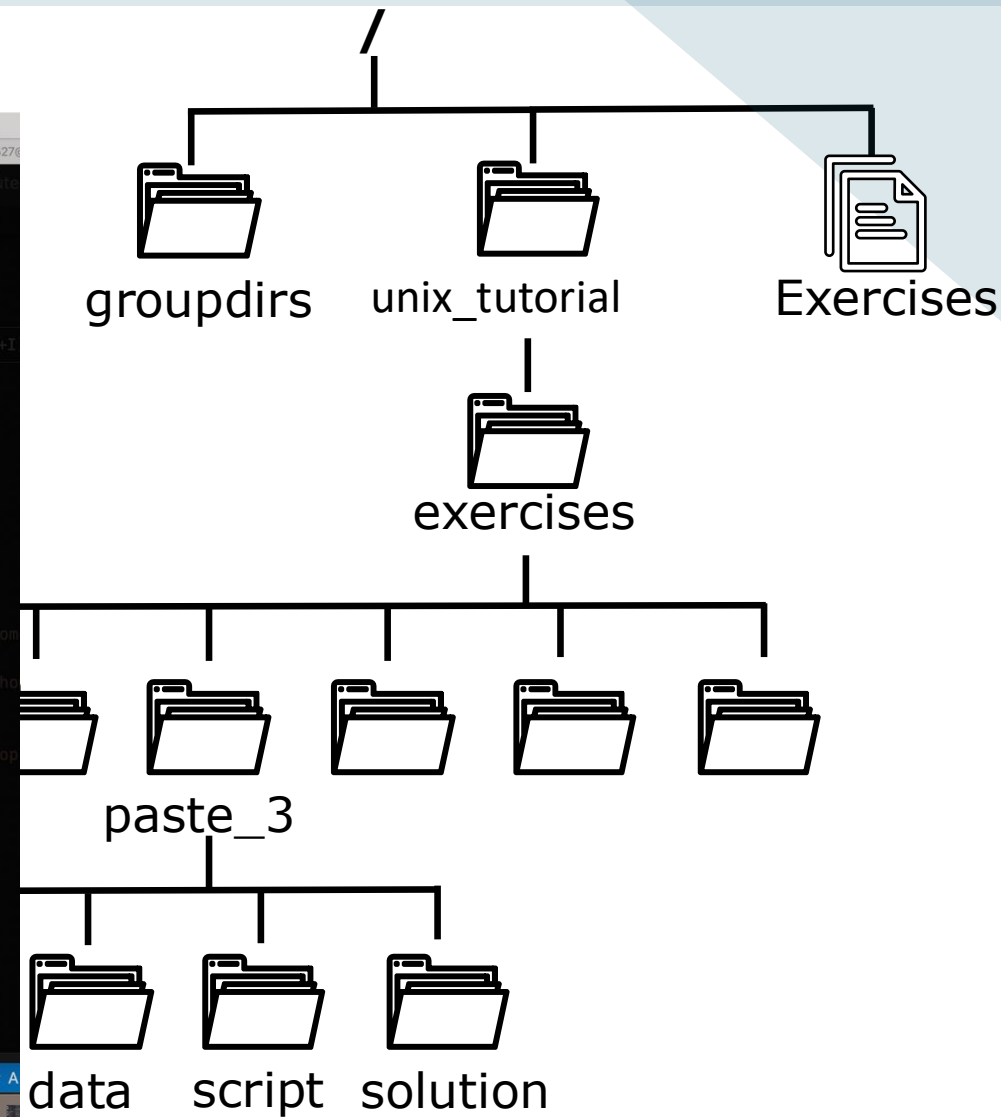
Server

Current
path
(working dir)

Prompt

File tree



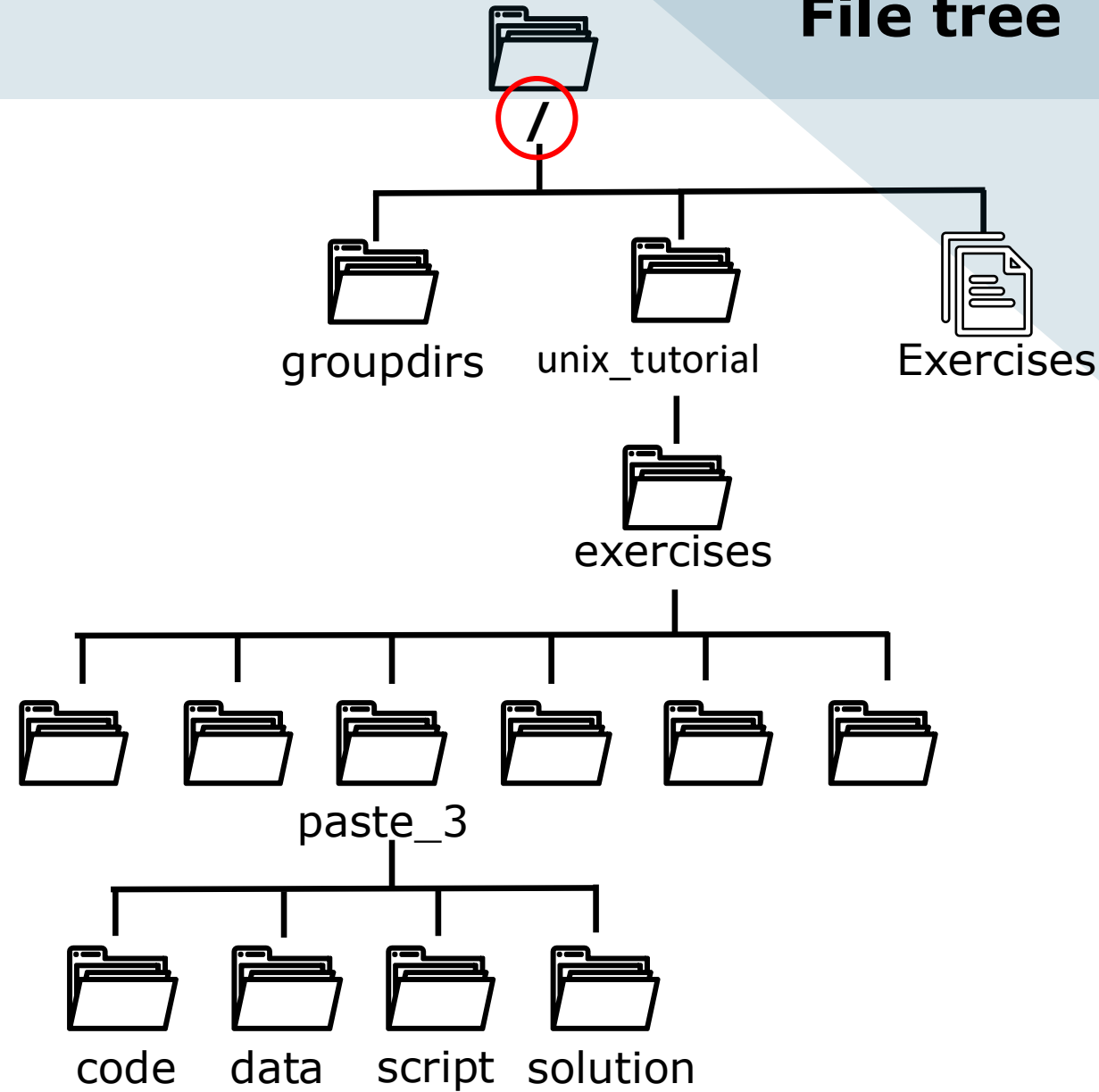


```
-- jbf527@mjoilnirgate01fl:/projects/mjoilnir1/people/jbf527/metagenomic_workshop -- ssh jbf527@mjoilnirgate.unicph.domain -- 137x34
jbf527@mjoilnirgate.unicph.domain -X jbf527@scarbpop05fl /jbf527/1.Rawfiles - bash /jbf527/1.Rawfiles - -zsh eindeer/1.Mapping - -zsh rse/exercises/linux - -zsh @a00730.science.domain jbf527
(/projects/mjoilnir1/apps/conda/py39) [jbf527@mjoilnirgate01fl:/projects/mjoilnir1/people/jbf527/metagenomic_workshop]$ tree
.
|-- Metaphlan
|   |-- final.merge.metaphlan4.txt
|   |-- GIL-1.bowtie2.out
|   |-- GIL-1.metaphlan4.txt
|   |-- HOR-1.bowtie2.out
|   |-- HOR-1.metaphlan4.txt
|   |-- merge.metaphlan4.txt
|   |-- metaphlan4.array.sh
|   |-- metaphlan4.squrt_scale.png
|   |-- modify.merge.metaphlan4.txt
|   |-- MSW-2765.bowtie2.out
|   |-- MSW-2765.metaphlan4.txt
|   |-- Samples.txt
|   |-- slurm-372588_1.out
|   |-- slurm-372588_2.out
|   `-- slurm-372588_3.out
-- MetaPhlAn-4.beta.1
|   |-- bioconda_recipe
|   |   `-- meta.yaml
|   |-- build
|   |   |-- bdist.linux-x86_64
|   |   `-- lib
|   |       `-- metaphlan
|   |           |-- __init__.py
|   |           |-- metaphlan_databases
|   |           |   |-- README.txt
|   |           |-- metaphlan.py
|   |           |-- strainphlan.py
|   |           `-- utils
|   |               |-- add_metadata_tree.py
|   |               |-- calculate_unifrac.R
|   |               `-- external_exec.py
|   `-- PopGenLinuxCourse.sh
|       2 ssh -Y -A -t <KU_user>@ssh-bio-stud.science.ku.dk ssh -Y -A -t popgen-bio
|       3
|       4 ssh -Y -A -t <KU_user>@a00730.science.domain ssh -Y -A -t popgen-bio
|       5
|       6 ssh -Y -A -t jbf527@a00730.science.domain ssh -Y -A -t popgen-bio
|       7
|       8 cp groupdirs/SCIENCE-BIO-Popgen_Course/exercises/linux/Exercises.zip .
|       9
|      10 ssh -Y -A -t <KU_user>@ssh-bio-stud.science.ku.dk ssh -Y -A -t popgen-bio
|      11
|      12 ## Advanced -- Taken from Just Bash Course
|      13 scp <KU_user>@ssh-bio-stud.science.ku.dk:~/groupdirs/SCIENCE-BIO-Popgen_Course/
|      14   exercises/linux/Advanced/Just_Bash_It_Exercises.html ~/Downloads
|      15
|      16 google-chrome -a Just_Bash_It_Exercises.html
|      17
|      18 rsync -rvt --progress Just-Bash-It-june_2023.zip jbf527@a00730.science.domain:/ho
|      19   jbf527/groupdirs/SCIENCE-BIO-Popgen_Course/exercises/linux/Advanced/
|      20 rsync -rvt --progress Just_Bash_It_Exercises.html jbf527@a00730.science.domain:/h
|      21   jbf527/groupdirs/SCIENCE-BIO-Popgen_Course/exercises/linux/Advanced/
|      22
|      23 rsync -rvt --progress jbf527@mjoilnirhead01fl.unicph.domain:/projects/mjoilnir1/peo
|      24   jbf527/backup_202* .
|      25
|      26 Solution
|      27
|      28 APTINYDYTES PATAGONICUS
|      29
```

File tree

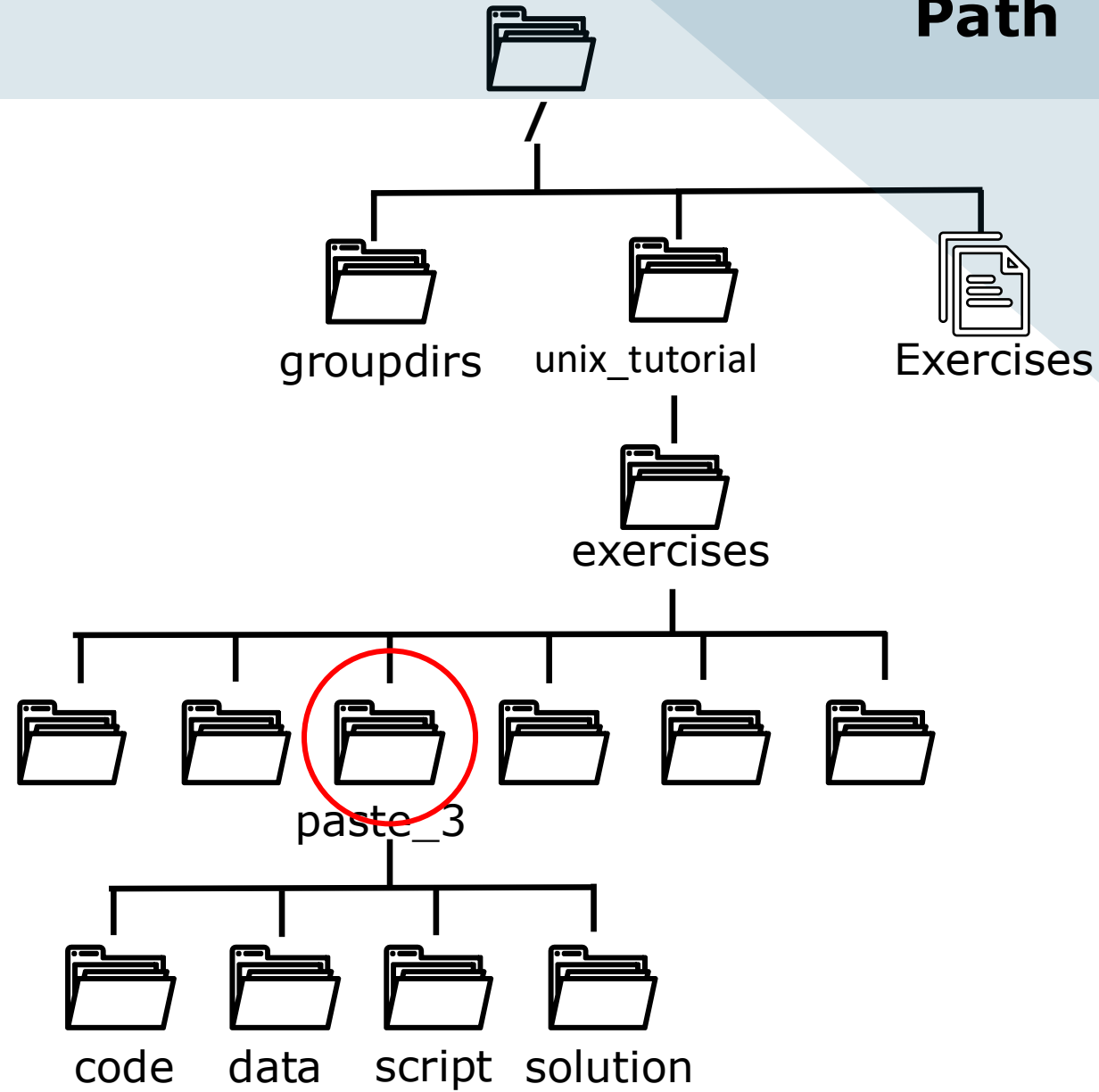
Branching and hierarchical structure

The topmost place is called the "root"




```
$ pwd
```

```
/unix_tutorial/exercises/paste_3
```



```
$ pwd
```

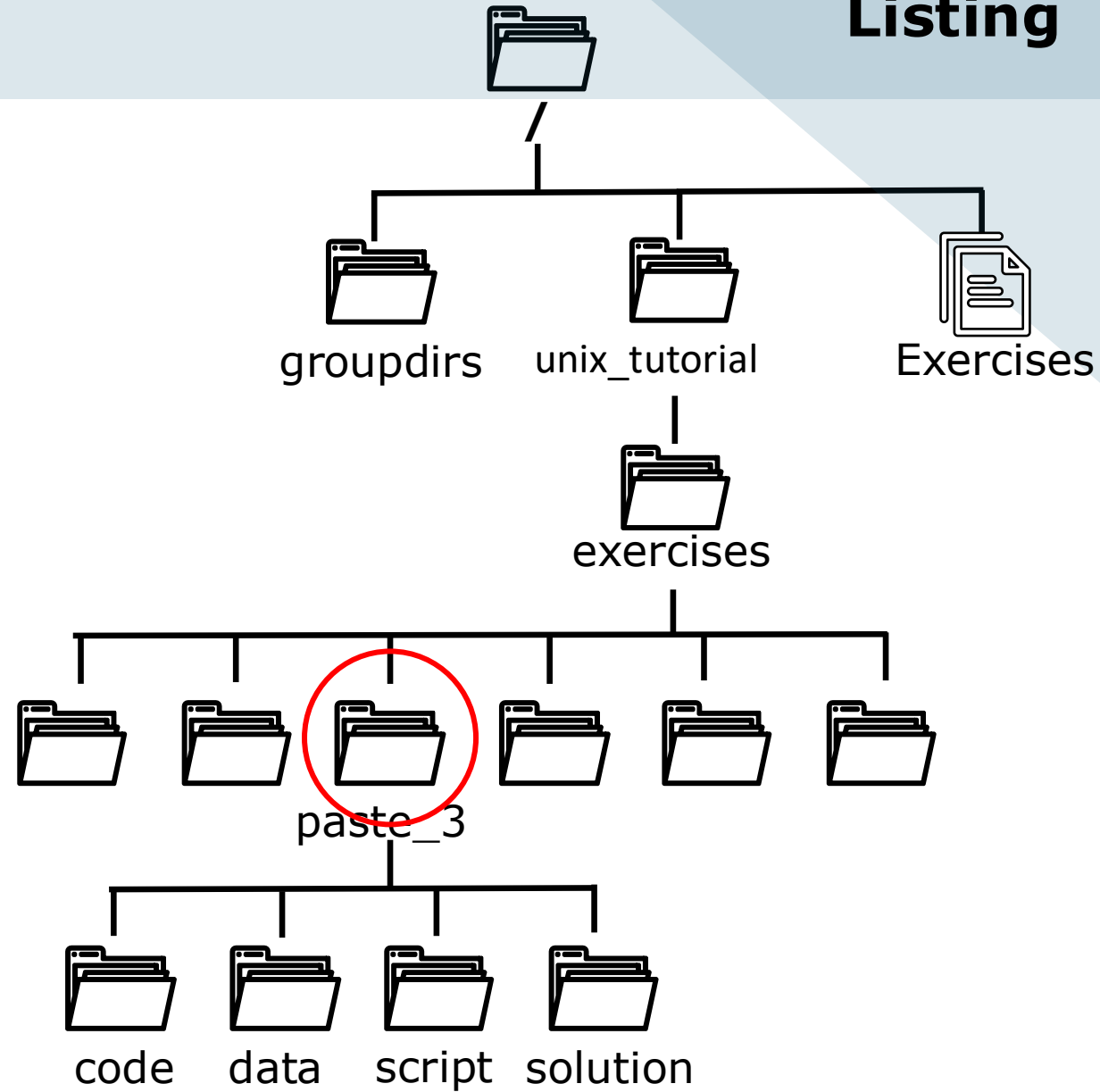
```
/unix_tutorial/exercises/paste_3
```

```
$ ls paste_3
```

```
code  
data  
script  
solution
```

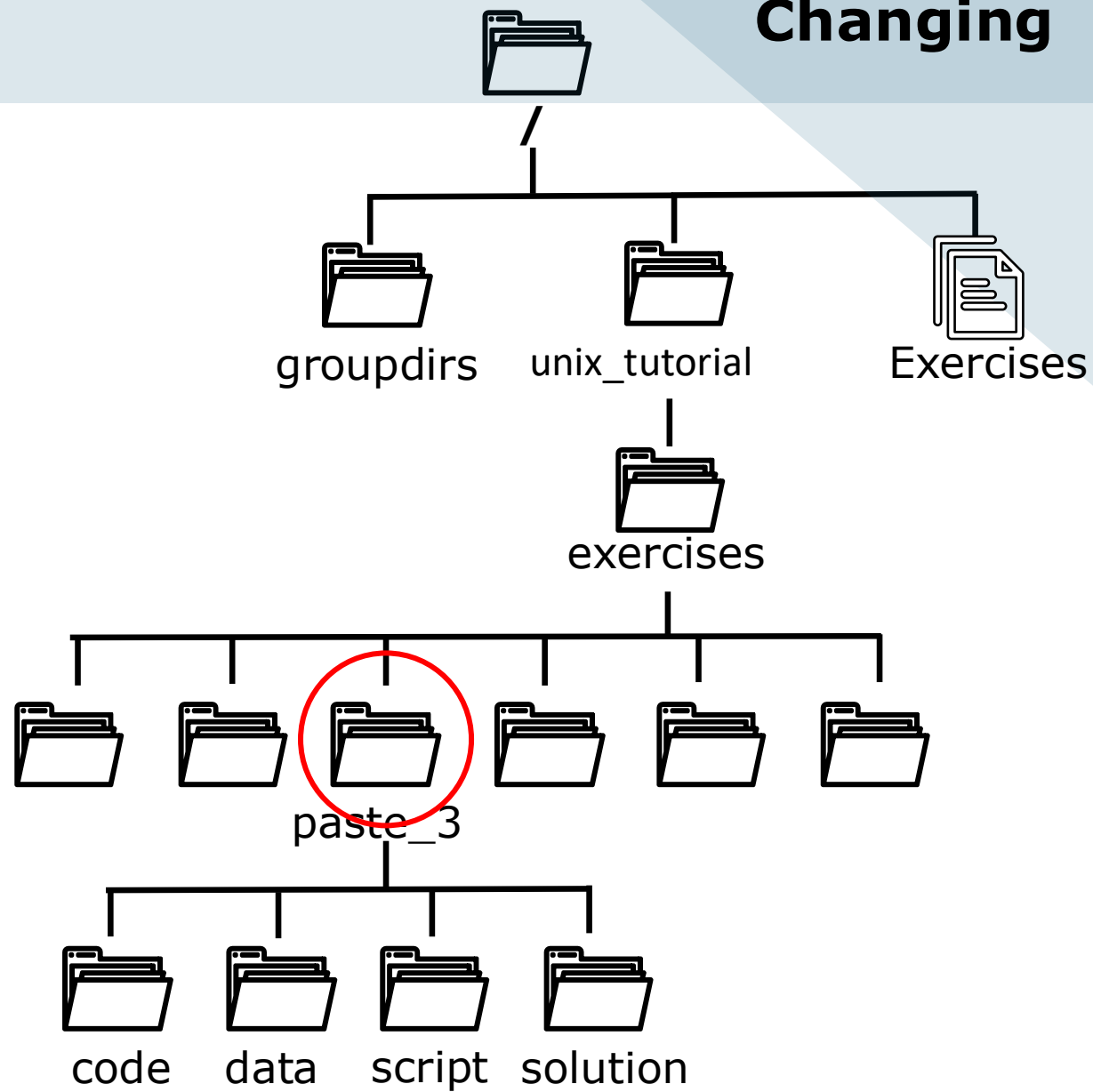
Many useful options:

- l
- t
- h



Changing

```
$ cd ..
```

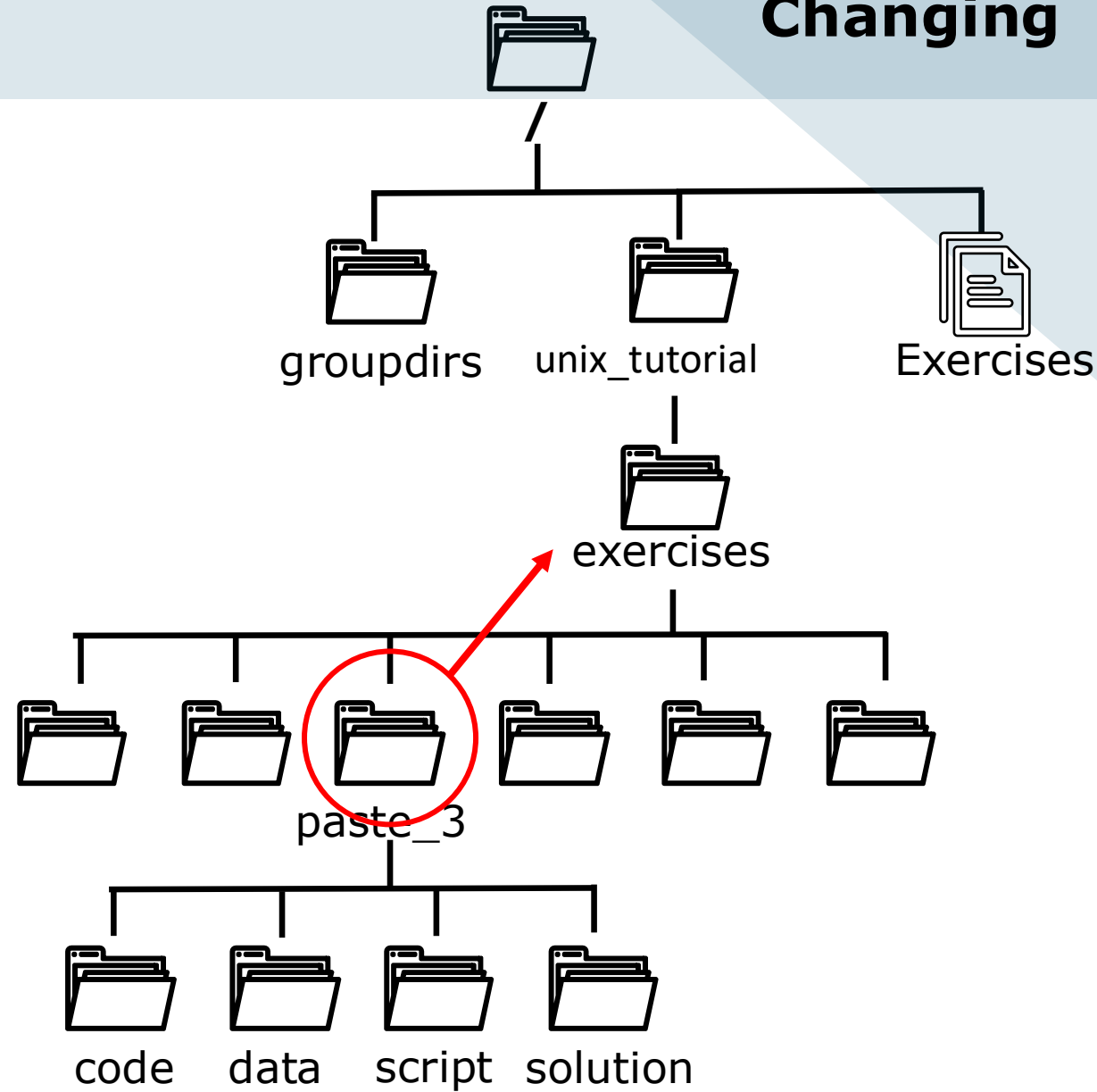


Changing

```
$ cd ..
```

```
$ pwd
```

```
/unix_tutorial/exercises
```

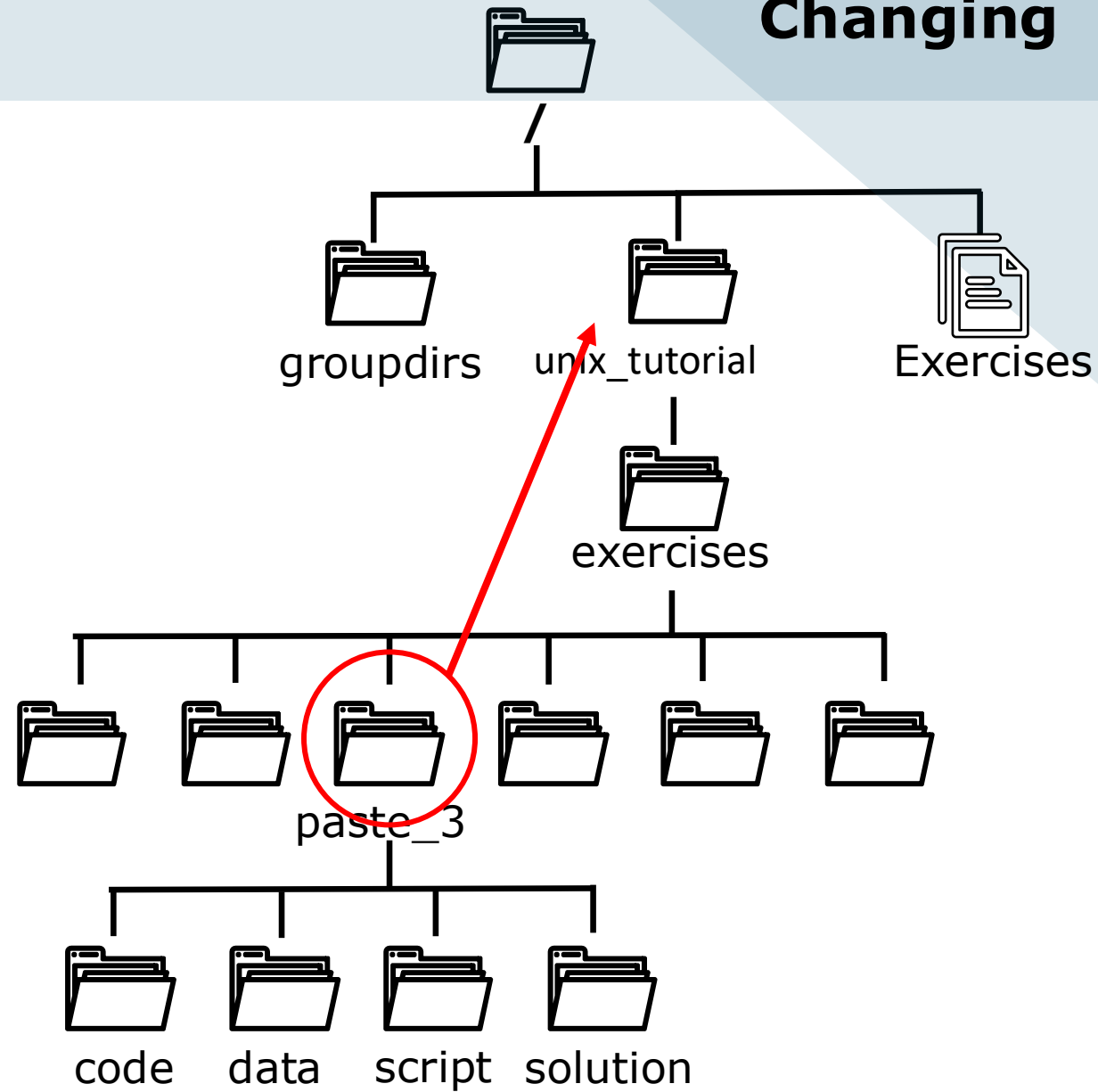


Changing

```
$ cd ../../
```

```
$ pwd
```

```
/unix_tutorial
```

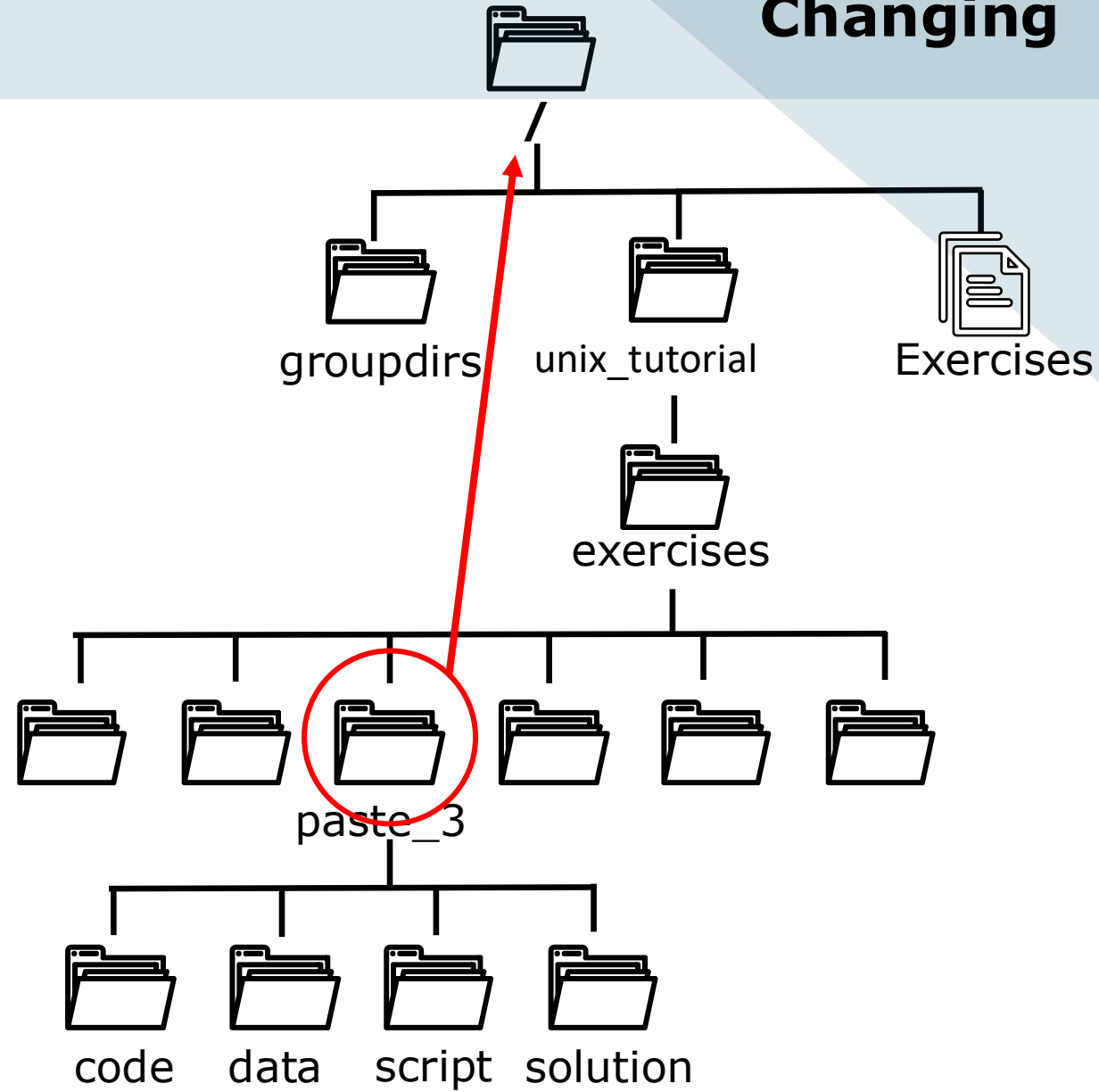


Changing

```
$ cd
```

```
$ pwd
```

```
/
```

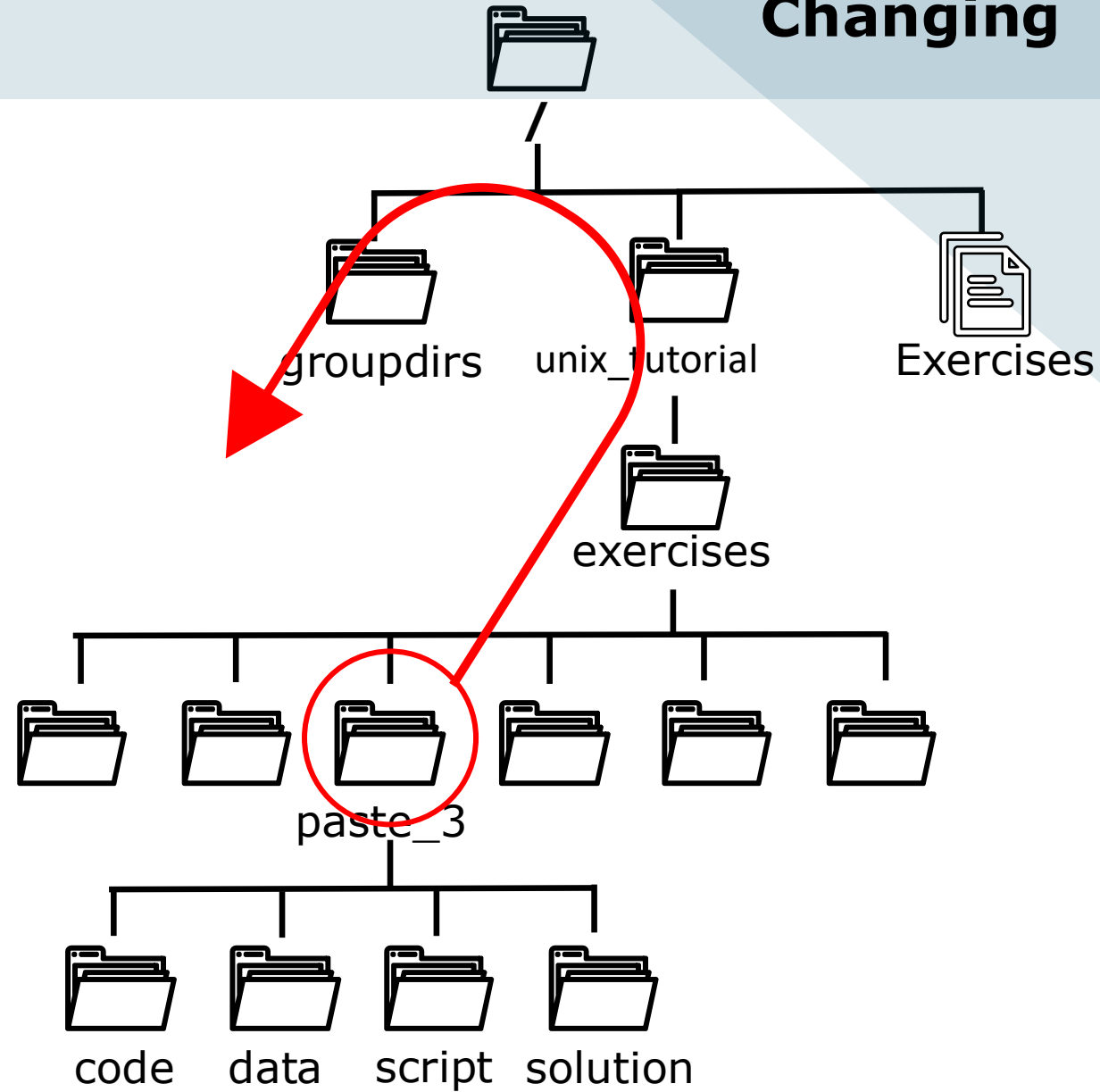


Changing

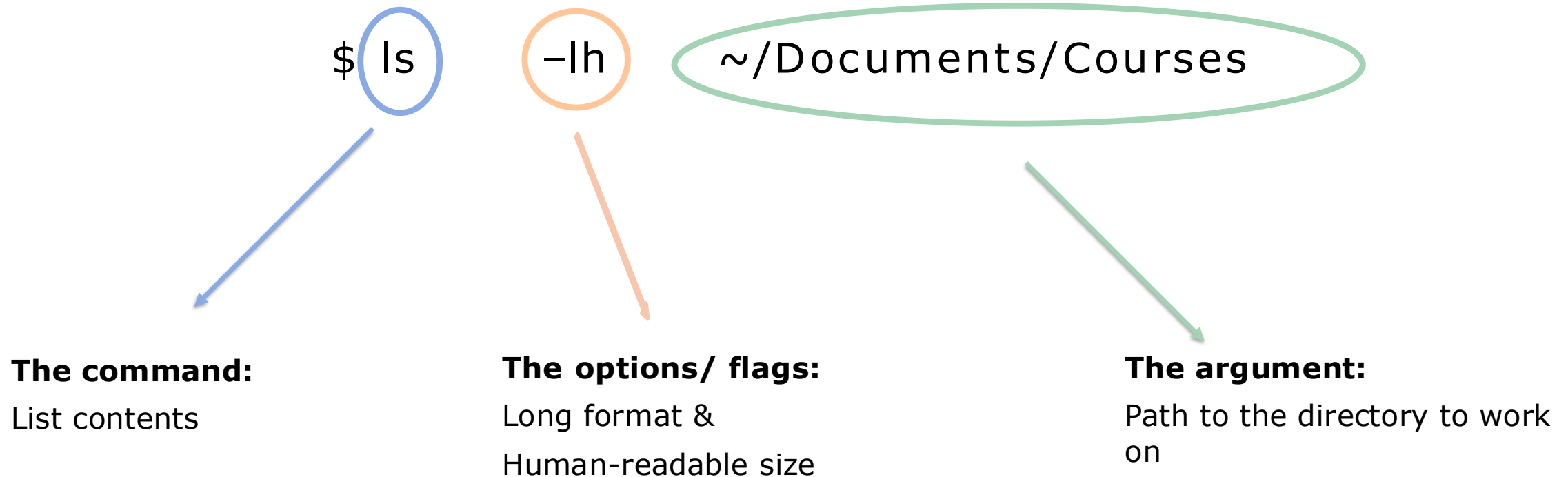
```
$ cd ../../../../groupdirs/dragons
```

```
$ pwd
```

```
/groupdirs/dragons
```



We interact with the computer by issuing commands to it. You have already tried some like **cd**, **ls**, **pwd**.
Some commands can stand by themselves like **pwd**, but often they have **arguments** and **options/flags**:



Whenever we put a **space** it means we are proceeding to new part of the command.

This is why file and directory names **cannot contain spaces**, they will be misinterpreted!

Commands options

```
bash-3.2$ ssh -Y -A -t jbf527@a00730.science.domain
jbf527@a00730.science.domain's password:
Last login: Tue Jan 28 15:08:10 2025 from 10.203.162.36
*
* Welcome to popgen-bio.
* Supported by UNICPH IT https://it.ku.dk/english/
*
* Notice: General system maintenance every 2nd thursday of each month 23:00-4:00
*
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file or directory
[jbf527@popgen-bio ~]$ tree
bash: tree: command not found...
[jbf527@popgen-bio ~]$ ls
Exercises.zip groupdirs unix_tutorial
[jbf527@popgen-bio ~]$ ls -l
total 472
-rwx----- 1 jbf527 sec-sci-users 134579 Jan 21 14:39 Exercises.zip
lrwxrwxrwx 1 jbf527 sec-sci-users 25 Jan 16 12:11 groupdirs -> /science/groupdirs/jbf527
drwxr-xr-x 3 jbf527 sec-sci-users 27 Jan 30 2023 unix_tutorial
[jbf527@popgen-bio ~]$ ls -lh
total 472K
-rwx----- 1 jbf527 sec-sci-users 132K Jan 21 14:39 Exercises.zip
lrwxrwxrwx 1 jbf527 sec-sci-users 25 Jan 16 12:11 groupdirs -> /science/groupdirs/jbf527
drwxr-xr-x 3 jbf527 sec-sci-users 27 Jan 30 2023 unix_tutorial
[jbf527@popgen-bio ~]$
[jbf527@popgen-bio ~]$ ls -lhtr
total 472K
drwxr-xr-x 3 jbf527 sec-sci-users 27 Jan 30 2023 unix_tutorial
lrwxrwxrwx 1 jbf527 sec-sci-users 25 Jan 16 12:11 groupdirs -> /science/groupdirs/jbf527
-rwx----- 1 jbf527 sec-sci-users 132K Jan 21 14:39 Exercises.zip
[jbf527@popgen-bio ~]$
```

\$ ls

\$ ls -l

\$ ls -lh

\$ ls -lhtr

Commands are case-sensitive

Note: Computers are dumb – case, spelling, and spacing matters!

i.e. test.txt $\neq$ Test.txt

Ls $\neq$ ls

ls -l $\neq$ ls -L

ls -l $\neq$ ls --l

ss -l $\neq$ ls -l

ls -l /home/user/Hello $\neq$ ls -l /home/user/hello

etc.

Often, checking your case, spelling, and spacing will solve most of your problems!

```
~ -- jbf527@a00730:~ -- bash -- 137x32
[jbf527@popgen-bio ~]$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
-a, --all                do not ignore entries starting with .
-A, --almost-all        do not list implied . and ..
--author                with -l, print the author of each file
-b, --escape             print C-style escapes for nongraphic characters
--block-size=SIZE        scale sizes by SIZE before printing them; e.g.,
                        '--block-size=M' prints sizes in units of
                        1,048,576 bytes; see SIZE format below
-B, --ignore-backups     do not list implied entries ending with ~
-c                      with -lt: sort by, and show, ctime (time of last
                        modification of file status information);
                        with -l: show ctime and sort by name;
                        otherwise: sort by ctime, newest first
-C                      list entries by columns
--color[=WHEN]          colorize the output; WHEN can be 'never', 'auto',
                        or 'always' (the default); more info below
-d, --directory          list directories themselves, not their contents
-D, --dired              generate output designed for Emacs' dired mode
-f                      do not sort, enable -aU, disable -ls--color
-F, --classify           append indicator (one of */=>@|) to entries
--file-type             likewise, except do not append '*'
--format=WORD            across -x, commas -m, horizontal -x, long -l,
                        single-column -1, verbose -l, vertical -C
--full-time             like -l --time-style=full-iso
-g                      like -l, but do not list owner
--group-directories-first
                        group directories before files;
```

\$ ls --help

\$ man ls


```
$ rm
```

This command removes forever. There is no recovery. **Careful!**

-r -> to remove whole directories

```
$ <command> *
```

Runs the command in every file in the directory

You can specify which files to use:

```
$ <command> *.txt
```

When things go wrong

```
$ cd paste_3
```

```
-bash: cd: paste_3: No such file or directory
```

```
$ Unicorn here.txt
```

```
-bash: Unicorn: command not found
```

```
(/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$ ll
total 20360
drwxr-xr-x. 3 jbf527 comp-prj-mjolnir1 29 Jul 8 2024 3.genetic_map_canfam31
drwxr-xr-x. 20 jbf527 comp-prj-mjolnir1 461 Dec 9 15:10 Africa
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 25876879 Jan 24 2024 Axiom_K9HDSNPA_Annotation.r1_3.bed
drwxr-xr-x. 5 jbf527 comp-prj-mjolnir1 371 Feb 10 2023 Danish_dogs
drwxr-xr-x. 5 jbf527 comp-prj-mjolnir1 93 Jan 2 17:35 Ducks_Samplix
drwxrwxrwx. 5 jbf527 comp-prj-mjolnir1 82 Jan 21 10:28 Elephants
drwxr-xr-x. 95 jbf527 comp-prj-mjolnir1 16275 Sep 24 18:22 HistoricalTEST
drwxr-xr-x. 39 jbf527 comp-prj-mjolnir1 4966 Jan 2 17:49 HumanAUS
drwxr-xr-x. 8 jbf527 comp-prj-mjolnir1 777 Dec 18 2022 aDNA_workshop
drwxr-xr-x. 1 jbf527 comp-prj-mjolnir1 90 Aug 9 2022 angsdput.arg
drwxr-xr-x. 1 jbf527 comp-prj-mjolnir1 1740 Apr 22 2024 backup_2024
drwxr-xr-x. 16 jbf527 comp-prj-mjolnir1 385 Jan 21 10:23 bin
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 11308758 Jan 25 2024 chip2.txt
drwxr-xr-x. 4 jbf527 comp-prj-mjolnir1 671 Jun 8 2023 frantk_workshop
drwxrwxr--. 31 jbf527 comp-prj-mjolnir1 999 Dec 6 15:55 jackals
drwxr-xr-x. 4 jbf527 comp-prj-mjolnir1 84 Dec 7 2022 metagenomic_workshop
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 3276 Jan 25 2023 paleomix_Awolves.error
drwx-----. 9 jbf527 comp-prj-mjolnir1 346 Jan 9 13:21 rawdata_NM
drwxrwxrwx. 12 jbf527 comp-prj-mjolnir1 290 Jan 8 14:59 refbackup_2024
drwx-----. 17 jbf527 comp-prj-mjolnir1 1214 Jan 14 14:59 rhino
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 70 Jun 17 2024 rint
drwx-----. 6 jbf527 comp-prj-mjolnir1 1587 Jan 8 14:59 scripts
-r-xr-x. 4 jbf527 comp-prj-mjolnir1 1207 Jun 20 2024 wolf_Africa
/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$ cd paste_3
-bash: cd: paste_3: No such file or directory
(/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$ Unicorn here.txt
-bash: Unicorn: command not found
(/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$
```

1. Do not panic!
2. Read the error message
3. Try to correct your error
4. Google or read the help page

Tab is your friend



Break

Prerequisites to run a program

1. The program must be on your system (i.e., computer, server).
2. The program (& files) must be executable. (Permission?)
3. Specify the interpreter to use & its location. (What?, How?)
4. Specify the location of the program on your system. (Where?)
 1. Current directory same as the program.
 2. Specify entire path of the program.
 3. The program's path is stored in the \$PATH environment variable.

When things go wrong

```
$ cd paste_3
```

```
-bash: cd: paste_3: No such file or directory
```

```
$ Unicorn here.txt
```

```
-bash: Unicorn: command not found
```

```
(/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$ ll
total 20360
drwxr-xr-x. 3 jbf527 comp-prj-mjolnir1 29 Jul 8 2024 3.genetic_map_canfam31
drwxr-xr-x. 20 jbf527 comp-prj-mjolnir1 461 Dec 9 15:10 Africa
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 25876879 Jan 24 2024 Axiom_K9HDSNPA_Annotation.r1_3.bed
drwxr-xr-x. 5 jbf527 comp-prj-mjolnir1 371 Feb 10 2023 Danish_dogs
drwxr-xr-x. 5 jbf527 comp-prj-mjolnir1 93 Jan 2 17:35 Ducks_Samplix
drwxrwxrwx. 5 jbf527 comp-prj-mjolnir1 82 Jan 21 10:28 Elephants
drwxr-xr-x. 95 jbf527 comp-prj-mjolnir1 16275 Sep 24 18:22 HistoricalTEST
drwxr-xr-x. 39 jbf527 comp-prj-mjolnir1 4966 Jan 2 17:49 HumanAUS
drwxr-xr-x. 8 jbf527 comp-prj-mjolnir1 777 Dec 18 2022 aDNA_workshop
drwxr-xr-x. 1 jbf527 comp-prj-mjolnir1 90 Aug 9 2022 angsdput.arg
drwxr-xr-x. 1 jbf527 comp-prj-mjolnir1 1740 Apr 22 2024 backup_2024
drwxr-xr-x. 16 jbf527 comp-prj-mjolnir1 385 Jan 21 10:23 bin
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 11308758 Jan 25 2024 chip2.txt
drwxr-xr-x. 4 jbf527 comp-prj-mjolnir1 671 Jun 8 2023 frantk_workshop
drwxrwxr--. 31 jbf527 comp-prj-mjolnir1 999 Dec 6 15:55 jackals
drwxr-xr-x. 4 jbf527 comp-prj-mjolnir1 84 Dec 7 2022 metagenomic_workshop
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 3276 Jan 25 2023 paleomix_Awolves.error
drwx-----. 9 jbf527 comp-prj-mjolnir1 346 Jan 9 13:21 rawdata_NM
drwxrwxrwx. 12 jbf527 comp-prj-mjolnir1 290 Jan 8 14:59 refbackup_2024
drwx-----. 17 jbf527 comp-prj-mjolnir1 1214 Jan 14 14:59 rhino
-rw-r--r--. 1 jbf527 comp-prj-mjolnir1 70 Jun 17 2024 rint
drwx-----. 6 jbf527 comp-prj-mjolnir1 1587 Jan 8 14:59 scripts
-r-xr-x. 4 jbf527 comp-prj-mjolnir1 1207 Jun 20 2024 wolf_Africa
/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$ cd paste_3
-bash: cd: paste_3: No such file or directory
(/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$ Unicorn here.txt
-bash: Unicorn: command not found
(/projects/mjolnir1/apps/conda/py39) ^ [jbf527@mjolnirgate01fl:/projects/mjolnir1/people/jbf527]$
```

1. Do not panic!
2. Read the error message
3. Try to correct your error
4. Google or read the help page

Tab is your friend

Prerequisites to run a program

1. The program must be on your system (i.e., computer, server).
2. The program (& files) must be executable. (Permission?)
3. Specify the interpreter to use & its location. (What?, How?)
4. Specify the location of the program on your system. (Where?)
 1. Current directory same as the program.
 2. Specify entire path of the program.
 3. The program's path is stored in the \$PATH environment variable.

Access rights/permissions to execute a program

- Each file / directory / program has its access rights
- Get overview of permissions by using 'ls -l'

```
[pbr316@popgen-bio ~]$ ls -l
total 120
l----- 1 pbr316 sec-sci-users 25 May 13 2019 groupdirs -> /science/groupdirs/pbr316
l----- 1 pbr316 sec-sci-users 22 May 13 2019 hdrive -> /science/hdrive/pbr316
-rw----- 1 pbr316 sec-sci-users 31 Feb 1 12:29 my first shell script.sh
```

```

|      |      |      |      |
| #links |      | bytesize |      |
permissions owner group date/time of filename
                        last modification

```

drwxrwxrwx

```

|_|_|_|_|
| | | |
|_ user |_ others
filetype group
```

Function
read
write
execute

Example: directory

- list files in the directory
- delete or move files in the directory
- access files in the directory

- `chmod -options <mode> <filename> [↵]`

```
chmod a+x my_first_shell_script.sh [↵]
```

```
ls -l [↵]
```

```
chmod g=r my_first_shell_script.sh [↵]
```

```
ls -l [↵]
```

```
chmod o-w my_first_shell_script.sh [↵]
```

```
ls -l [↵]
```

Prerequisites to run a program

1. The program must be on your system (i.e., computer, server).
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3. **Specify the interpreter to use & its location. (What?, How?)**
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Specify the interpreter to use

- `#!/` is called shebang
- Specifies the rest of the line as the directory of the interpreter
- Implies the programming language used in the files remainder
- `/bin/bash` as interpreter of `my_first_shell_script.sh`

```
GNU nano 2.0.6 File: my_first_shell_script.sh
#!/bin/bash
ls -l
cd ..
ls -l
```



Other languages using the command line

```
$ bash my_script.sh
```

```
$ python align_reads.py
```

```
$ Rscript deSEQ_cancer_Data.R
```

- We can use other languages in the interpreter
- Languages like R and python – they need to be installed in the server/computer

Prerequisites to run a program

1. The program must be on your system (i.e., computer, server).
2. The program (& files) must be executable. (Permission?)
3. Specify the interpreter to use & its location. (What?, How?)
4. Specify the location of the program on your system. (Where?)
 1. Current directory same as the program.
 2. Specify entire path of the program.
 3. The program's path is stored in the \$PATH environment variable.

Specify the location of the program

- Current directory same as the program:

```
[pbr316@popgen-bio ~]$ bash ./my_first_shell_script.sh
```

- Specify the entire path:

```
[pbr316@popgen-bio groupdirs]$ bash /home/pbr316/my_first_shell_script.sh
```

- Program's path is part of the \$PATH environment variable:

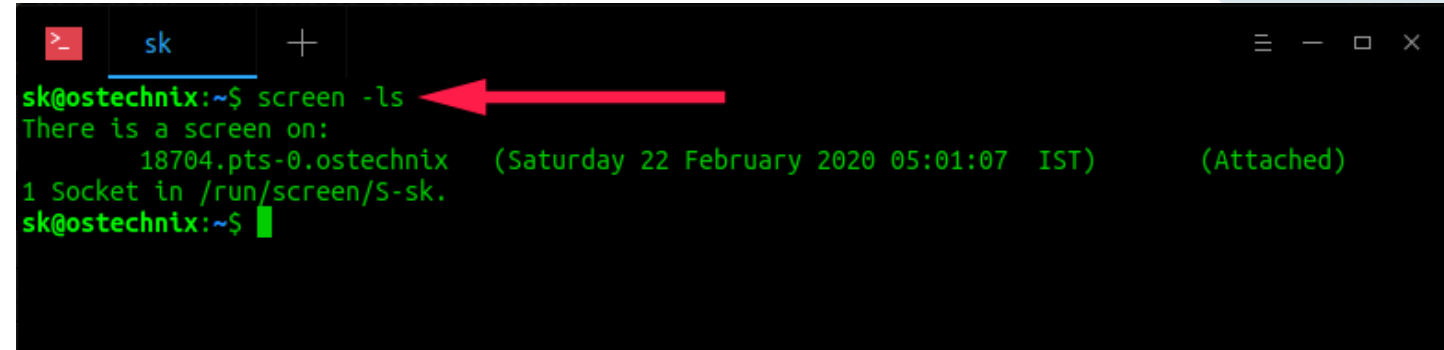
```
[pbr316@popgen-bio ~]$ bash my_first_shell_script.sh
```

- \$PATH is an environmental variable
- includes a list of predefined directories
- system checks if program exists in any of those locations
- otherwise: "command not found"

For long-running tasks

SSH session is terminated

- Terminal multiplexer
- Creates virtual windows

A terminal window with a dark background and green text. The prompt is 'sk@ostechnix:~\$'. The command 'screen -ls' has been entered. The output shows 'There is a screen on:' followed by '18704.pts-0.ostechnix (Saturday 22 February 2020 05:01:07 IST) (Attached)' and '1 Socket in /run/screen/S-sk.'. A red arrow points to the 'screen -ls' command.

```
sk@ostechnix:~$ screen -ls
There is a screen on:
      18704.pts-0.ostechnix  (Saturday 22 February 2020 05:01:07 IST)      (Attached)
1 Socket in /run/screen/S-sk.
sk@ostechnix:~$
```

\$ screen

Description

Start a new screen

List running screens

Attach to a running screen

Attach to a running screen with name

Detach a running screen

Kill a running screen

Command

screen

screen -ls

screen -x

screen -r <screen_name>

screen -d <screen_name>

screen -X -S [screen] kill

Within Screen

detach

Create new window

Kill current window

Command

Ctrl-a d

Ctrl-a c

Ctrl-a k

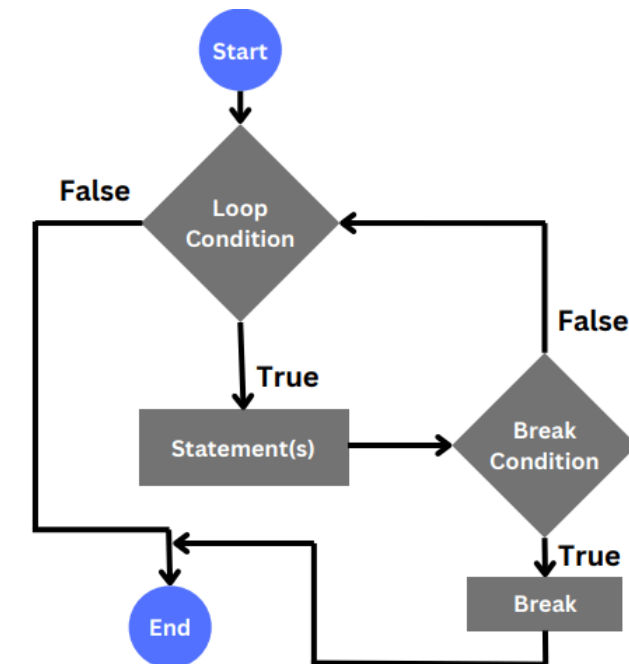
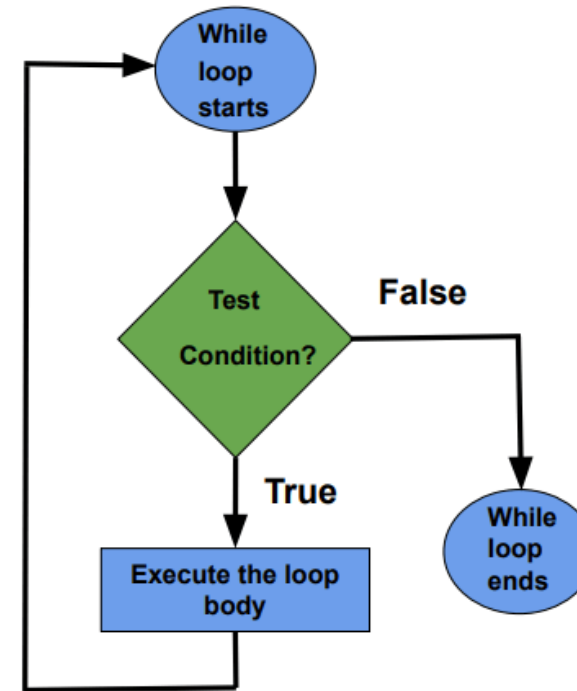
For many tasks, use loops:

- **For**
- **While**
- **Until**
- **If**
- **Scripts**

```
GNU nano 6.2 /usr/local/bin/demo.sh *
#!/bin/bash

# Define your array
countries=("India" "France" "United Kingdom")

# Loop through the array
for country in "${countries[@]}"
do
    echo "$country"
done
```



- The Linux command line replaces graphical interfaces with a unified, universal syntax – ‘One to rule them all.’
- Free, versatile and widespread use
- Allows automatization of almost all processes
- Allows interaction with a multitude of programs in bioinformatics
- Indispensable for running analyses on large datasets

Task: enter the terminal

Browser

Open any browser and log in to Jupyter notebook:

Log on to server

<https://emily.popgen.dk/hub/login>

Log on to server

Username: <Your own>

Password: <You have to define it – PLEASE DO NOT FORGET>

Jupyter notebook interface

 Logout Control Panel

Files Running Clusters

Select items to perform actions on them. Upload New ▾ ↺

☐ 0 ▾

 /

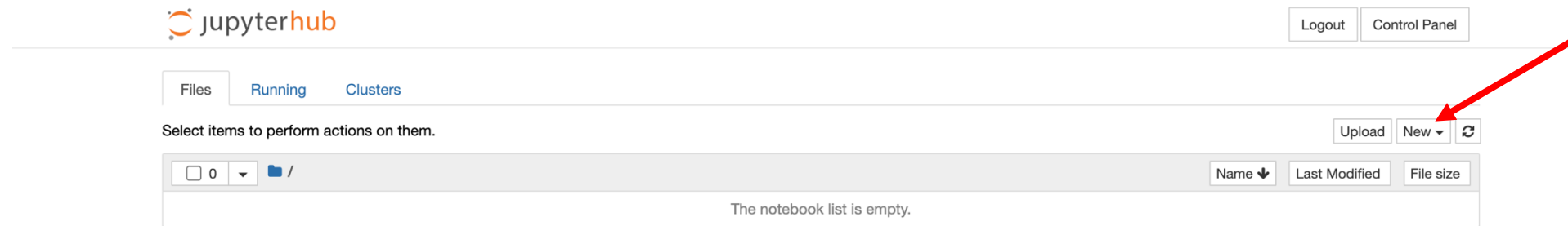
Name ▾

Last Modified

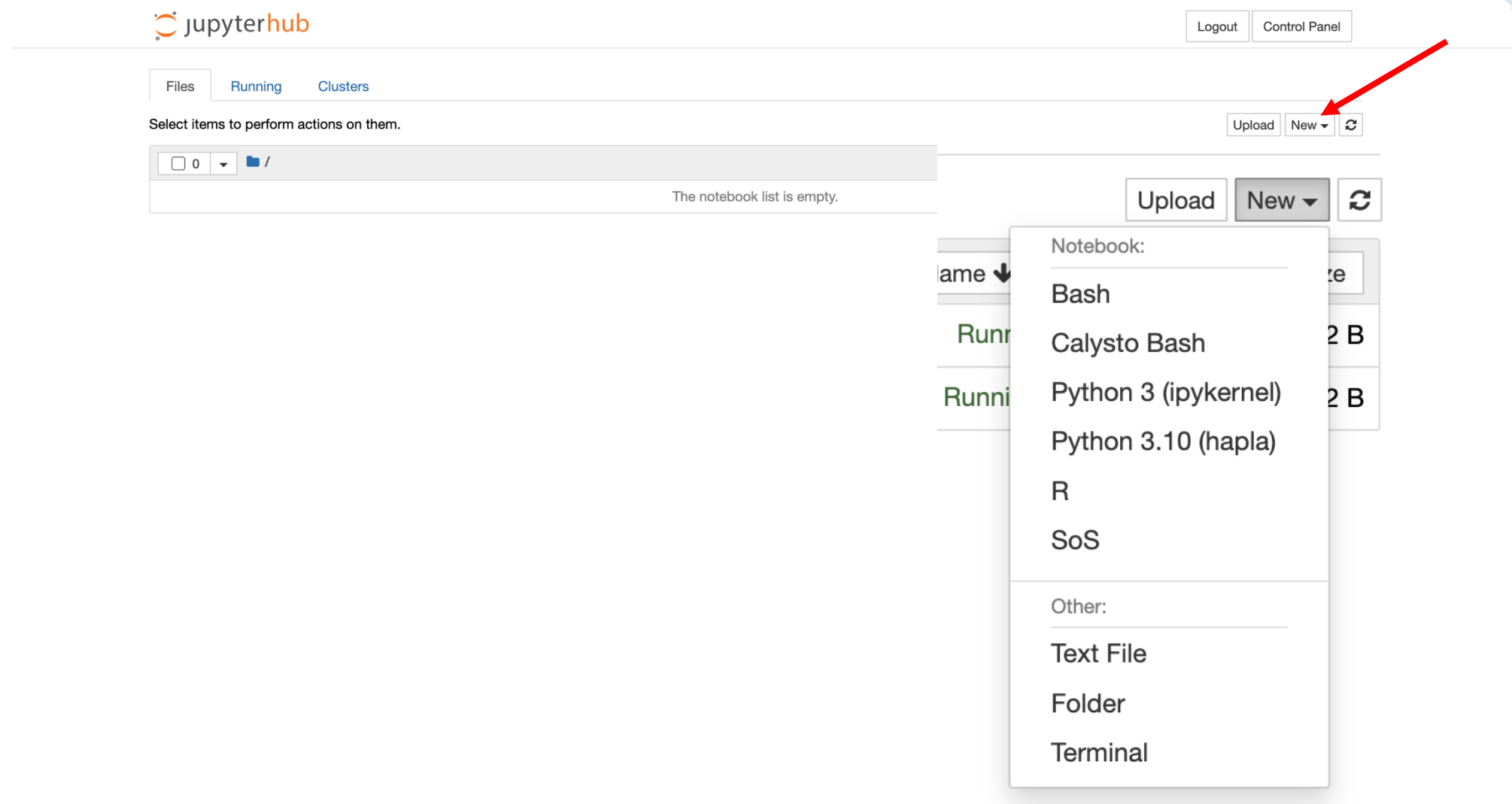
File size

The notebook list is empty.

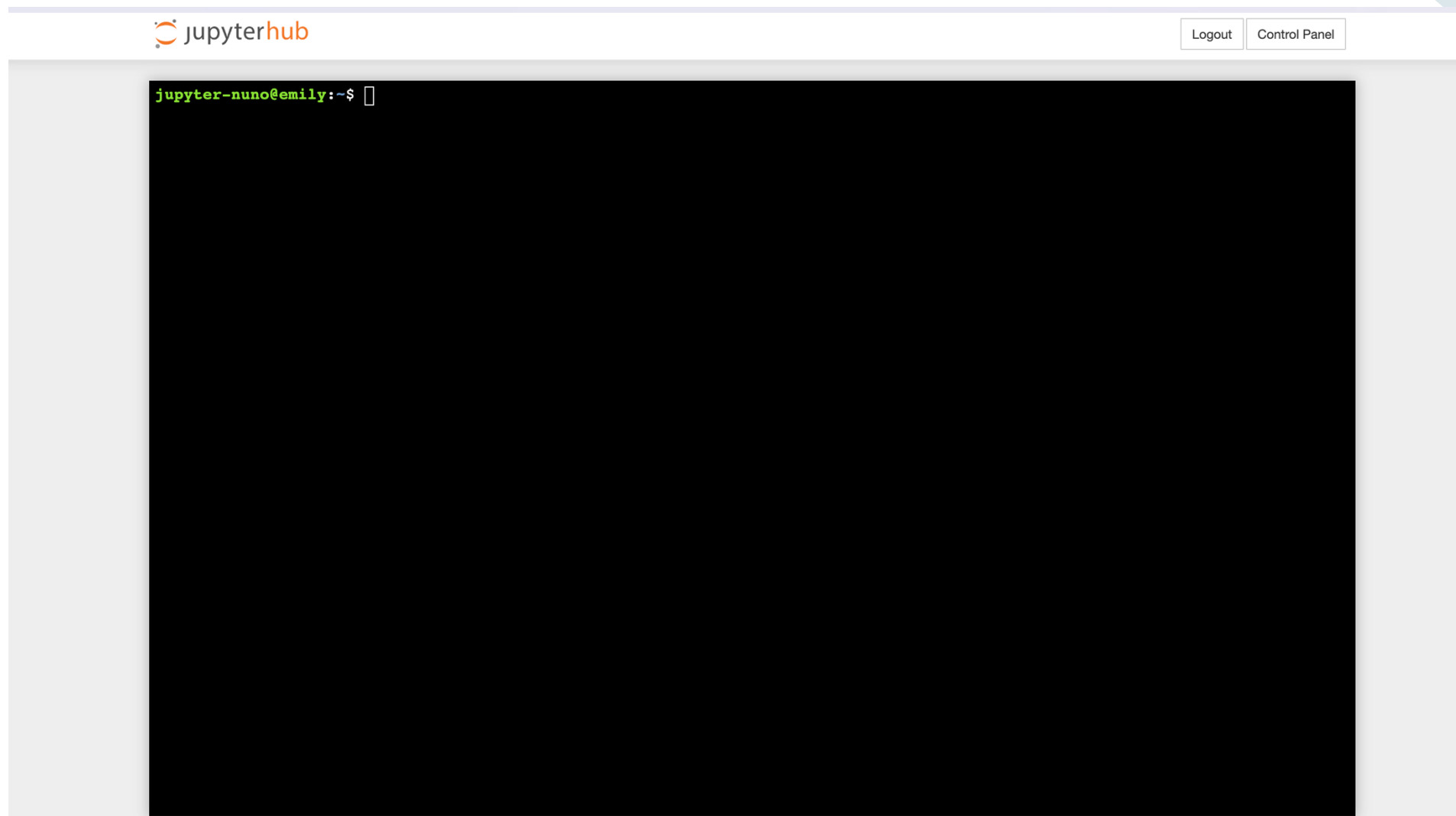
Jupyter notebook interface



Jupyter notebook interface



Jupyter notebook interface



Jupyter notebook interface

 jupyterhub

Logout

Control Panel

```
jupyter-nuno@emily:~$ ls
'Intro to Linux.ipynb'
jupyter-nuno@emily:~$ ll
total 76
drwxr-x---  8 jupyter-nuno jupyter-nuno  4096 Aug 12 15:35 ./
drwxr-xr-x 329 root        root        20480 Aug 10 14:41 ../
-rw-----  1 jupyter-nuno jupyter-nuno 10041 Aug 12 15:35 .bash_history
-rw-r--r--  1 jupyter-nuno jupyter-nuno   220 Jan  6  2022 .bash_logout
-rw-r--r--  1 jupyter-nuno jupyter-nuno  3771 Jan  6  2022 .bashrc
drwxr-xr-x  3 jupyter-nuno jupyter-nuno  4096 Aug  1  2025 .cache/
-rw-r--r--  1 jupyter-nuno jupyter-nuno   959 Aug 12 15:33 'Intro to Linux.ipynb'
drwxr-xr-x  2 jupyter-nuno jupyter-nuno  4096 Aug 12 15:35 .ipynb_checkpoints/
drwxr-xr-x  3 jupyter-nuno jupyter-nuno  4096 Aug  3  2025 .ipython/
drwxr-xr-x  2 jupyter-nuno jupyter-nuno  4096 Aug  3  2025 .jupyter/
drwxr-xr-x  3 jupyter-nuno jupyter-nuno  4096 Aug  1  2025 .local/
-rw-r--r--  1 jupyter-nuno jupyter-nuno   807 Jan  6  2022 .profile
drwxr-xr-x  6 jupyter-nuno jupyter-nuno  4096 Aug 12 15:28 .sos/
jupyter-nuno@emily:~$ cd paste3
bash: cd: paste3: No such file or directory
jupyter-nuno@emily:~$ Unicorn
Command 'Unicorn' not found, did you mean:
  command 'unicorn' from deb unicorn (6.0.0-2build1)
Try: apt install <deb name>
jupyter-nuno@emily:~$
```

```
for object to mirror  
mirror_mod.mirror_object  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True
```

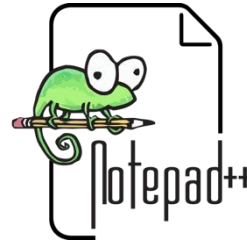
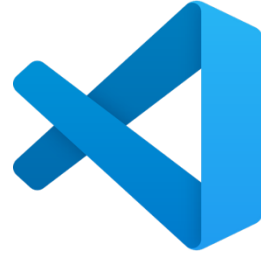
```
selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob))  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
print("please select exactly
```

-- OPERATOR CLASSES ----

```
types.Operator):  
X mirror to the selected  
object.mirror_mirror_x"  
mirror X"
```

- **Text editor**

- Visual Studio Code
- Notepad++
- Sublime
- Atom
- Notepad



- **For Linux command help:**

- ``man <command>``
- Linux reference guides online

- **Google, StackOverflow (or maybe ChatGPT or Perplexity) is your best friend:**

- Most of the time there's no need to reinvent the wheel – someone out there may already have a solution to your question!

- **Practice makes perfect!**

Instructions are on Absalon:

1. Log on to jupyter

<https://emily.popgen.dk/hub/login>

2. Exercise instructions

<https://github.com/popgenDK/courses/tree/main/kenya2026/exercises/Day1/course/popgenmsc26/exercises/linux/Exercises.zip>

https://github.com/Nuno-GMartins/bash_tutorial

3. Extra exercises using the terminal

[Just-Bash-It-june_2023.zip](#)

[Just_Bash_It_Exercises.html](#)

[/course/popgenmsc26/exercises/linux/Advanced/](#)



Credit/ original version: https://github.com/krother/bash_tutorial18

Credit/ original version: https://github.com/Center-for-Health-Data-Science/Just-Bash-It/tree/june_2023